MATH 535: LINEAR ALGEBRA

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1. August 24, 2026

1.1. **Fields.**

Definition 1 (Binary operation)**.** *A* binary operation *on a set S is a function*

$$F\colon S \times S \longrightarrow S.$$

Thus, for every $x, y \in S$, the value $F(x, y)$ is again an element of S. We usually write $x \cdot y$ or $x + y$ in place of $F(x, y)$.

Example 1. *Addition on $\mathbb{Z}$ is a binary operation: the map*

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \qquad (m, n) \longmapsto m + n$$

takes every pair of integers to an integer.

Definition 2. *Let F be a binary operation on S. A subset $T \subseteq S$ is* closed under F *if*

$$F(x, y) \in T$$

whenever $x, y \in T$.

Example 2. *The positive integers $\mathbb{Z}_{>0}$ are closed under addition. The odd integers are not closed under addition, but they are closed under multiplication.*

Definition 3 (Field)**.** *Let $\mathbb{F}$ be a set equipped with two binary operations, called* addition *and* multiplication*. The sum and product of $a, b \in \mathbb{F}$ are denoted by $a + b$ and ab, respectively. Suppose that for every $a, b, c \in \mathbb{F}$,*

(i) $a + b = b + a$;
(ii) $(a + b) + c = a + (b + c)$;
(iii) $ab = ba$;
(iv) $a(bc) = (ab)c$;
(v) $a(b + c) = ab + ac$.

Suppose further that

(vi) *$\mathbb{F}$ contains an additive identity 0 and a multiplicative identity 1, so that*

$$a + 0 = a \qquad \text{and} \qquad 1a = a$$

for every $a \in \mathbb{F}$;
(vii) $1 \neq 0$;
(viii) *every $a \in \mathbb{F}$ has an additive inverse $-a \in \mathbb{F}$ such that $a + (-a) = 0$;*

(ix) every nonzero $a \in \mathbb{F}$ has a multiplicative inverse $a^{-1} \in \mathbb{F}$ such that $aa^{-1} = 1$.

Then $\mathbb{F}$ is called a field.

Example 3. *The integers $\mathbb{Z}$ do not form a field: for example, 2 has no multiplicative inverse in $\mathbb{Z}$. The sets $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$, with their usual operations, are fields.*

There are also finite fields. For example,

$$\mathbb{F}_2 = \{0, 1\} \qquad \text{and} \qquad \mathbb{F}_3 = \{0, 1, 2\},$$

where the elements are the possible remainders upon division by 2 and 3, respectively. Addition and multiplication in $\mathbb{F}_3$ are given by the following tables:

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

More generally, if p is prime, then

$$\mathbb{F}_p = \{0, 1, \ldots, p-1\},$$

with addition and multiplication performed modulo p, is a field.

Lemma 1. *If a is an element of a field $\mathbb{F}$, then $a \cdot 0 = 0$.*

Proof. Since $0 + 0 = 0$, distributivity gives

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding the additive inverse of $a \cdot 0$ to both sides yields $0 = a \cdot 0$. □

2. August 26, 2026

2.1. **Further properties of fields.** The preceding lecture showed that $a \cdot 0 = 0$ for every element a of a field. We now obtain the zero-product property.

Proposition 1 (Zero-product property)**.** *Let $a, b \in \mathbb{F}$. If $ab = 0$, then either $a = 0$ or $b = 0$.*

Proof. Suppose that $ab = 0$ and $a \neq 0$. Since $\mathbb{F}$ is a field, a^{-1} exists. Therefore

$$0 = a^{-1}0 = a^{-1}(ab) = (a^{-1}a)b = 1b = b.$$

Thus $a = 0$ or $b = 0$. □

2.2. **Characteristic of a field.** For a positive integer n and $a \in \mathbb{F}$, write

$$na = \underbrace{a + a + \cdots + a}_{n \text{ terms}}.$$

Definition 4 (Characteristic)**.** *Let $\mathbb{F}$ be a field. If $n1 = 0$ for some positive integer n, then the least such n is called the* characteristic *of $\mathbb{F}$, denoted* $\mathrm{char}(\mathbb{F})$*. If $n1 \neq 0$ for every positive integer n, then $\mathbb{F}$ has characteristic zero.*

The fields $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ have characteristic zero. For every prime p, the field $\mathbb{F}_p$ has characteristic p. The rational function field $\mathbb{F}_p(t)$ is an infinite field of characteristic p.

Proposition 2. *If $\mathbb{F}$ has positive characteristic r, then r is prime.*

Proof. We have $r \neq 1$, since $1 \neq 0$. Suppose that r is composite, so

$$r = st$$

for integers s, t satisfying $1 < s, t < r$. Distributivity gives

$$(s1)(t1) = (st)1 = r1 = 0.$$

By the zero-product property, either $s1 = 0$ or $t1 = 0$. Both alternatives contradict the minimality of r. Hence r is prime. □

Proposition 3. *If* $\mathrm{char}(\mathbb{F}) = p > 0$*, then*

$$pa = 0$$

for every $a \in \mathbb{F}$.

Proof. Since $p1 = 0$, associativity and distributivity imply

$$pa = p(1a) = (p1)a = 0a = 0.$$

□

Proposition 4. *Every finite field has positive characteristic.*

Proof. Let $\mathbb{F}$ be finite. The sequence

$$1, \quad 2 \cdot 1, \quad 3 \cdot 1, \quad \dots$$

cannot consist of distinct elements. Hence there are positive integers $m > n$ such that

$$m1 = n1.$$

Adding the additive inverse of $n1$ to both sides gives

$$(m - n)1 = 0.$$

Thus $\mathbb{F}$ has positive characteristic. □

2.3. **Algebraically closed fields.**

Definition 5 (Algebraically closed field)**.** *A field $\mathbb{F}$ is* algebraically closed *if every nonconstant polynomial $f \in \mathbb{F}[x]$ has a root in $\mathbb{F}$.*

The field $\mathbb{Q}$ is not algebraically closed, because $x^2 - 2$ has no root in $\mathbb{Q}$. The field $\mathbb{R}$ is not algebraically closed, because $x^2 + 1$ has no real root. By the fundamental theorem of algebra, $\mathbb{C}$ is algebraically closed.

Proposition 5. *No finite field is algebraically closed.*

Proof. Let

$$\mathbb{F} = \{a_1, \dots, a_q\}$$

be finite. Consider the nonconstant polynomial

$$f(x) = 1 + \prod_{i=1}^{q} (x - a_i).$$

For each $i = 1, \dots, q$,

$$f(a_i) = 1 \neq 0.$$

Thus f has no root in $\mathbb{F}$, so $\mathbb{F}$ is not algebraically closed. □

2.4. **Vectors and vector spaces.** The spaces $\mathbb{R}^2$, $\mathbb{R}^3$, and, more generally,

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \ldots, x_n \in \mathbb{R} \right\}$$

are the motivating examples of vector spaces.

Definition 6 (Vector space)**.** *Let $\mathbb{F}$ be a field. A* vector space over $\mathbb{F}$ *is a set V equipped with two operations,*

$$V \times V \longrightarrow V, \qquad (u, v) \longmapsto u + v,$$

and

$$\mathbb{F} \times V \longrightarrow V, \qquad (\alpha, v) \longmapsto \alpha v,$$

satisfying the following axioms for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{F}$:

(V1) $u + (v + w) = (u + v) + w$;
(V2) $u + v = v + u$;
(V3) there is an element $0 \in V$ *such that* $v + 0 = 0 + v = v$;
(V4) for every $v \in V$*, there is an element* $-v \in V$ *such that* $v + (-v) = (-v) + v = 0$;
(V5) $1v = v$;
(V6) $(\alpha\beta)v = \alpha(\beta v)$;
(V7) $\alpha(u + v) = \alpha u + \alpha v$;
(V8) $(\alpha + \beta)v = \alpha v + \beta v$.

The elements of V are called *vectors*, and the elements of $\mathbb{F}$ are called *scalars*.

3. August 28, 2026

3.1. **Examples of vector spaces.**

Example 4 (Coordinate spaces)**.** *Let $\mathbb{F}$ be a field. The set*

$$\mathbb{F}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \ldots, x_n \in \mathbb{F} \right\}$$

is a vector space over $\mathbb{F}$ *under componentwise addition and scalar multiplication:*

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \qquad \alpha \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{bmatrix}.$$

In particular, $\mathbb{R}^n$ *is a vector space over* $\mathbb{R}$ *and* $\mathbb{C}^n$ *is a vector space over* $\mathbb{C}$*.*

Example 5 (Matrix spaces)**.** *The set* $M_{m\times n}(\mathbb{F})$ *of* $m \times n$ *matrices with entries in* $\mathbb{F}$ *is a vector space over* $\mathbb{F}$*. Addition and scalar multiplication are defined entrywise:*

$$[a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}], \qquad \alpha[a_{ij}] = [\alpha a_{ij}].$$

Example 6 (Polynomial spaces)**.** *Let*

$$\mathcal{P}_n(\mathbb{F}) = \{a_0 + a_1 t + \cdots + a_n t^n : a_0, \ldots, a_n \in \mathbb{F}\}.$$

This is the vector space over $\mathbb{F}$ *of polynomials of degree at most* n*. Its operations are*

$$\begin{aligned}(a_0 + a_1 t + \cdots + a_n t^n) + (b_0 + b_1 t + \cdots + b_n t^n) \\ = (a_0 + b_0) + (a_1 + b_1)t + \cdots + (a_n + b_n)t^n,\end{aligned}$$

and

$$\alpha(a_0 + a_1 t + \cdots + a_n t^n) = \alpha a_0 + \alpha a_1 t + \cdots + \alpha a_n t^n.$$

3.2. **Linear combinations and spanning sets.**

Definition 7 (Linear combination and span)**.** *Let* V *be a vector space over* $\mathbb{F}$ *and let* $v_1, \ldots, v_p \in V$*. A* linear combination *of these vectors is a vector of the form*

$$\alpha_1 v_1 + \cdots + \alpha_p v_p, \qquad \alpha_1, \ldots, \alpha_p \in \mathbb{F}.$$

The set of all their linear combinations is called their span*:*

$$\operatorname{span}\{v_1, \ldots, v_p\} = \{\alpha_1 v_1 + \cdots + \alpha_p v_p : \alpha_1, \ldots, \alpha_p \in \mathbb{F}\}.$$

A linear combination is trivial *if all of its coefficients are zero.*

In $\mathbb{R}^2$, let

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then

$$\operatorname{span}\{e_1, e_2\} = \mathbb{R}^2,$$

because

$$\begin{bmatrix} x \\ y \end{bmatrix} = xe_1 + ye_2.$$

If

$$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

then $\{e_1, e_2, v\}$ also spans $\mathbb{R}^2$, since

$$\begin{bmatrix} x \\ y \end{bmatrix} = xe_1 + ye_2 + 0v.$$

The additional vector is unnecessary, which motivates the question of redundancy in a spanning set.

Definition 8 (Spanning set)**.** *A collection $\{v_1, \ldots, v_p\} \subseteq V$ is a* spanning set *for V if every $v \in V$ can be written as a linear combination*

$$v = \alpha_1 v_1 + \cdots + \alpha_p v_p.$$

Equivalently,

$$\operatorname{span}\{v_1, \ldots, v_p\} = V.$$

3.3. **Linear independence.**

Definition 9 (Linear independence)**.** *A collection $\{v_1, \ldots, v_p\} \subseteq V$ is* linearly independent *if*

$$\alpha_1 v_1 + \cdots + \alpha_p v_p = 0$$

implies

$$\alpha_1 = \cdots = \alpha_p = 0.$$

In other words, only the trivial linear combination of $v_1, \ldots, v_p$ equals zero. A collection that is not linearly independent is called linearly dependent*.*

Example 7. *Consider the vectors*

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

in $\mathbb{R}^3$*. To test linear independence, solve*

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

This gives the system

$$\begin{cases} \alpha + 2\beta + 3\gamma = 0, \\ \alpha + \gamma = 0, \\ \beta + \gamma = 0. \end{cases}$$

Taking $\gamma = 1$ *gives* $\alpha = \beta = -1$*, so*

$$-v_1 - v_2 + v_3 = 0.$$

Thus the vectors are linearly dependent; equivalently, $v_3 = v_1 + v_2$*.*

The following result precisely characterizes redundancy in a finite collection of vectors.

Proposition 6. *Let* $v_1, \dots, v_p \in V$*. The collection* $\{v_1, \dots, v_p\}$ *is linearly dependent if and only if one of its vectors is a linear combination of the others. Equivalently, there is an index* k *and scalars* $\beta_j \in \mathbb{F}$ *such that*

$$v_k = \sum_{\substack{j=1 \\ j \neq k}}^{p} \beta_j v_j.$$

Proof. Suppose first that $\{v_1, \dots, v_p\}$ is linearly dependent. Then there are scalars $\alpha_1, \dots, \alpha_p$, not all zero, such that

$$\alpha_1 v_1 + \cdots + \alpha_p v_p = 0.$$

Choose k such that $\alpha_k \neq 0$. Since $\mathbb{F}$ is a field, α_k^{-1} exists, and we can solve for v_k:

$$v_k = -\sum_{\substack{j=1 \\ j\neq k}}^{p} \alpha_k^{-1}\alpha_j v_j.$$

Thus v_k is a linear combination of the other vectors.

Conversely, suppose that for some k,

$$v_k = \sum_{\substack{j=1 \\ j\neq k}}^{p} \beta_j v_j.$$

Moving all terms to one side gives

$$v_k - \sum_{\substack{j=1 \\ j\neq k}}^{p} \beta_j v_j = 0.$$

This is a nontrivial linear combination of $v_1, \ldots, v_p$, because the coefficient of v_k is $1 \neq 0$. Hence the collection is linearly dependent. □

4. August 31, 2026

4.1. **Bases and coordinates.**

Definition 10 (Basis)**.** *A collection of vectors $\{v_1, \ldots, v_n\}$ in a vector space V is a* basis *for V if it is linearly independent and spans V.*

The two conditions in the definition can be combined into a statement about representing vectors uniquely.

Proposition 7. *A collection $\{v_1, \ldots, v_n\}$ is a basis for V if and only if every vector $v \in V$ has a unique representation of the form*

$$v = a_1 v_1 + \cdots + a_n v_n, \qquad a_1, \ldots, a_n \in \mathbb{F}.$$

When $\{v_1, \ldots, v_n\}$ is a basis, the scalars $a_1, \ldots, a_n$ are called the coordinates *of v with respect to this basis.*

Proof. Suppose first that $\{v_1, \ldots, v_n\}$ is a basis. Since the collection spans V, every $v \in V$ has a representation

$$v = a_1 v_1 + \cdots + a_n v_n.$$

To prove uniqueness, suppose that

$$v = \widetilde{a}_1 v_1 + \cdots + \widetilde{a}_n v_n$$

is another such representation. Subtracting the two equations gives

$$(a_1 - \widetilde{a}_1)v_1 + \cdots + (a_n - \widetilde{a}_n)v_n = 0.$$

Because $\{v_1, \ldots, v_n\}$ is linearly independent,

$$a_i - \widetilde{a}_i = 0$$

for each i. Thus $a_i = \widetilde{a}_i$ for every i, and the representation is unique.

Conversely, suppose every $v \in V$ has a unique representation as a linear combination of $v_1, \ldots, v_n$. The existence of such a representation for every v shows that the collection spans V. Now suppose

$$a_1 v_1 + \cdots + a_n v_n = 0.$$

The zero vector also has the representation

$$0 = 0v_1 + \cdots + 0v_n.$$

Uniqueness therefore gives $a_1 = \cdots = a_n = 0$. Hence the collection is linearly independent, so it is a basis for V. □

Example 8 (The standard basis of $\mathbb{F}^n$)**.** *For $i = 1, \ldots, n$, let $e_i \in \mathbb{F}^n$ be the vector whose ith coordinate is 1 and whose other coordinates are 0. Thus*

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \qquad \ldots \qquad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Every vector in $\mathbb{F}^n$ has the unique representation

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n.$$

Consequently, $\{e_1, \dots, e_n\}$ *is a basis for* $\mathbb{F}^n$*, called its* standard basis*.*

Example 9 (The standard basis of $\mathcal{P}_n(\mathbb{F})$)**.** *Recall that* $\mathcal{P}_n(\mathbb{F})$ *is the vector space of polynomials over* $\mathbb{F}$ *of degree at most* n*. Each polynomial has a unique expression*

$$p(t) = a_0 + a_1 t + \cdots + a_n t^n.$$

Hence

$$\{1, t, t^2, \dots, t^n\}$$

is a basis for $\mathcal{P}_n(\mathbb{F})$*, called its standard basis.*

Proposition 8. *Every finite nonempty spanning set for a vector space contains a basis.*

Proof. Let $\{v_1, \dots, v_p\}$ span V. If this collection is linearly independent, then it is already a basis. Otherwise, by Proposition 6, some v_k is a linear combination of the other vectors. It follows that removing v_k does not change the span:

$$\operatorname{span}\{v_1, \dots, v_p\} = \operatorname{span}\{v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_p\}.$$

If the remaining collection is linearly independent, it is a basis. If not, repeat the same argument. The original spanning set is finite, so this process must terminate. The final collection still spans V and is linearly independent, and is therefore a basis contained in the original spanning set. □

4.2. **Linear transformations.** A *transformation*, also called a map or function, from a set X to a set Y is a rule $T\colon X \to Y$ that assigns to every $x \in X$ an element $T(x) \in Y$.

Definition 11 (Linear transformation)**.** *Let* V *and* W *be vector spaces over the same field* $\mathbb{F}$*. A transformation* $T\colon V \to W$ *is* linear *if*

(i) $T(u + v) = T(u) + T(v)$ *for all* $u, v \in V$*;*
(ii) $T(\alpha u) = \alpha T(u)$ *for all* $u \in V$ *and* $\alpha \in \mathbb{F}$*.*

Equivalently, $T\colon V \to W$ is linear if

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

for all $u, v \in V$ and $\alpha, \beta \in \mathbb{F}$.

Example 10 (Reflection)**.** *Define* $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ *by*

$$T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}.$$

Geometrically, T *reflects each point across the first coordinate axis. To check additivity, observe that*

$$\begin{aligned} T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) &= T\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 + y_1 \\ -x_2 - y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ -y_2 \end{bmatrix} \\ &= T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + T\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}. \end{aligned}$$

For $\alpha \in \mathbb{R}$*,*

$$\begin{aligned} T\left(\alpha\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) &= T\begin{bmatrix} \alpha x_1 \\ \alpha x_2 \end{bmatrix} \\ &= \begin{bmatrix} \alpha x_1 \\ -\alpha x_2 \end{bmatrix} \\ &= \alpha\begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} = \alpha T\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \end{aligned}$$

Thus T *is linear.*

Example 11 (The differentiation operator)**.** *Let* $n \geq 1$*,* $V = \mathcal{P}_n(\mathbb{F})$*, and* $W = \mathcal{P}_{n-1}(\mathbb{F})$*. Define*

$$T\colon V \longrightarrow W, \qquad T(p) = p'.$$

Since

$$(p+q)' = p' + q' \qquad \text{and} \qquad (\alpha p)' = \alpha p',$$

the differentiation operator T *is linear.*

5. September 2, 2026

5.1. **Linear transformations determined by a basis.** The following result shows that prescribing the values of a linear transformation on a basis completely determines the transformation.

Lemma 2 (Linear map lemma)**.** *Suppose that $\{v_1, \dots, v_n\}$ is a basis for V, and let $w_1, \dots, w_n$ be arbitrary vectors in W. Then there exists a unique linear transformation $T\colon V \to W$ such that*

$$T(v_k) = w_k$$

for each $k = 1, \dots, n$.

Proof. Every $v \in V$ has a unique representation

$$v = \sum_{i=1}^{n} c_i v_i.$$

Define

$$T(v) = \sum_{i=1}^{n} c_i w_i.$$

The uniqueness of the coordinates $c_1, \dots, c_n$ shows that this definition is well-defined. In particular, taking $v = v_k$ gives $T(v_k) = w_k$.

To prove that T is linear, write

$$u = \sum_{i=1}^{n} a_i v_i \qquad \text{and} \qquad v = \sum_{i=1}^{n} c_i v_i.$$

Then

$$\begin{aligned} T(u+v) &= T\left(\sum_{i=1}^{n} (a_i + c_i) v_i\right) \\ &= \sum_{i=1}^{n} (a_i + c_i) w_i \\ &= \sum_{i=1}^{n} a_i w_i + \sum_{i=1}^{n} c_i w_i \\ &= T(u) + T(v). \end{aligned}$$

Similarly, for every $\alpha \in \mathbb{F}$,

$$\begin{aligned} T(\alpha v) &= T\left(\sum_{i=1}^{n} \alpha c_i v_i\right) \\ &= \sum_{i=1}^{n} \alpha c_i w_i = \alpha \sum_{i=1}^{n} c_i w_i = \alpha T(v). \end{aligned}$$

Thus T is linear.

It remains to prove uniqueness. Suppose that $T' \colon V \to W$ is another linear transformation satisfying $T'(v_k) = w_k$ for every k. If $v = \sum_{i=1}^{n} c_i v_i$, then

$$T'(v) = \sum_{i=1}^{n} c_i T'(v_i) = \sum_{i=1}^{n} c_i w_i = T(v).$$

Hence $T' = T$. □

5.2. **Matrix representations of linear transformations.** An *ordered basis* is a basis whose vectors are listed in a fixed order. Let

$$\mathcal{B} = (v_1, \ldots, v_n) \qquad \text{and} \qquad \mathcal{C} = (w_1, \ldots, w_m)$$

be ordered bases for V and W, respectively. For $v = x_1 v_1 + \cdots + x_n v_n$, write

$$[v]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

for the coordinate vector of v with respect to $\mathcal{B}$.

Theorem 1 (Matrix representation theorem)**.** *Let V and W be vector spaces over $\mathbb{F}$ with ordered bases $\mathcal{B} = (v_1, \ldots, v_n)$ and $\mathcal{C} = (w_1, \ldots, w_m)$. For every linear transformation $T \colon V \to W$, there is a unique matrix*

$$A = [A_{ij}] = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \in M_{m \times n}(\mathbb{F})$$

such that

$$T(v_j) = \sum_{i=1}^{m} A_{ij} w_i$$

for each $j = 1, \ldots, n$. Conversely, every matrix $A \in M_{m\times n}(\mathbb{F})$ determines a unique linear transformation $T\colon V \to W$ through this formula.

Proof. Let $T\colon V \to W$ be linear. Because $\mathcal{C}$ is a basis for W, each vector $T(v_j)$ has a unique representation

$$T(v_j) = A_{1j}w_1 + \cdots + A_{mj}w_m = \sum_{i=1}^{m} A_{ij}w_i.$$

Use these coordinate vectors as the columns of A:

$$A = \begin{bmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{bmatrix}.$$

This constructs the required matrix. If $B = [B_{ij}]$ were another matrix with the same property, then for every j,

$$\sum_{i=1}^{m} B_{ij}w_i = T(v_j) = \sum_{i=1}^{m} A_{ij}w_i.$$

The uniqueness of coordinates with respect to $\mathcal{C}$ gives $B_{ij} = A_{ij}$ for every i and j. Hence $B = A$.

Conversely, let $A = [A_{ij}]$ be an arbitrary $m \times n$ matrix and define

$$\widetilde{w}_j = \sum_{i=1}^{m} A_{ij}w_i \qquad (j = 1, \ldots, n).$$

By the linear map lemma, there is a unique linear transformation $T\colon V \to W$ such that $T(v_j) = \widetilde{w}_j$ for every j. Thus A determines a unique linear transformation with the required property. □

If

$$v = x_1v_1 + \cdots + x_nv_n,$$

then linearity gives

$$\begin{aligned} T(v) &= \sum_{j=1}^{n} x_jT(v_j) \\ &= \sum_{j=1}^{n} x_j \sum_{i=1}^{m} A_{ij}w_i \end{aligned}$$

$$= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} A_{ij} x_j \right) w_i.$$

Consequently,

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

Thus multiplication by the matrix A computes the coordinates of the image of a vector from the coordinates of the original vector.

6. September 4, 2026

6.1. **Matrix–vector multiplication.** Let V and W be vector spaces over $\mathbb{F}$ with ordered bases

$$\mathcal{B} = (v_1, \ldots, v_n) \qquad \text{and} \qquad \mathcal{C} = (w_1, \ldots, w_m),$$

respectively, and let $T\colon V \to W$ be linear. Recall that the matrix representing T with respect to these bases is

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix},$$

where

$$T(v_j) = A_{1j} w_1 + \cdots + A_{mj} w_m.$$

Thus the jth column of A is the coordinate vector $[T(v_j)]_{\mathcal{C}}$.

Suppose that

$$v = x_1 v_1 + \cdots + x_n v_n, \qquad [v]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

By linearity,

$$\begin{aligned} [T(v)]_{\mathcal{C}} &= x_1 [T(v_1)]_{\mathcal{C}} + \cdots + x_n [T(v_n)]_{\mathcal{C}} \\ &= x_1 \begin{bmatrix} A_{11} \\ A_{21} \\ \vdots \\ A_{m1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} A_{1n} \\ A_{2n} \\ \vdots \\ A_{mn} \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} A_{11}x_1 + \cdots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n \end{bmatrix}.$$

This motivates the following definition.

Definition 12 (Matrix–vector product)**.** *Let $A = [A_{ij}] \in M_{m\times n}(\mathbb{F})$, and write its columns as $A_1, \ldots, A_n$. For*

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{F}^n,$$

define

$$Ax = x_1A_1 + \cdots + x_nA_n.$$

Equivalently, the kth coordinate of Ax is

$$(Ax)_k = \sum_{j=1}^{n} A_{kj}x_j, \qquad k = 1, \ldots, m.$$

With this notation, the coordinate formula for a linear transformation is

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

Example 12. *Matrix–vector multiplication can be computed either as a linear combination of the columns or one coordinate at a time. For instance,*

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 14 \\ 10 \end{bmatrix},$$

while

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + 2\cdot 2 + 3\cdot 3 \\ 4\cdot 1 + 5\cdot 2 + 6\cdot 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \end{bmatrix}.$$

6.2. **Finding matrix representations.**

Example 13. *Define* $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ *by*

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x - 5y \\ 7y \end{bmatrix}.$$

With respect to the standard bases, the columns of the representing matrix are

$$T(e_1) = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \qquad \text{and} \qquad T(e_2) = \begin{bmatrix} 2 \\ -5 \\ 7 \end{bmatrix}.$$

Hence

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -5 \\ 0 & 7 \end{bmatrix}.$$

Example 14 (Differentiation)**.** *Let* $n \geq 1$*, and let*

$$T\colon \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}), \qquad T(p) = p',$$

and use the ordered basis

$$\mathcal{B} = (1, t, t^2, \ldots, t^n)$$

for both the domain and codomain. Since

$$T(1) = 0, \qquad T(t) = 1, \qquad T(t^j) = jt^{j-1},$$

the matrix representing T *is*

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & n \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}.$$

6.3. **The vector space of linear maps.** Let V and W be vector spaces over $\mathbb{F}$. If $T\colon V \to W$ is linear and $\alpha \in \mathbb{F}$, define $\alpha T\colon V \to W$ pointwise by

$$(\alpha T)(v) = \alpha T(v).$$

This map is again linear. Indeed, for $u, v \in V$,

$$\begin{aligned}(\alpha T)(u+v) &= \alpha T(u+v)\\ &= \alpha\big(T(u)+T(v)\big)\\ &= (\alpha T)(u)+(\alpha T)(v),\end{aligned}$$

and, for $\beta \in \mathbb{F}$,

$$(\alpha T)(\beta u) = \alpha T(\beta u) = \alpha\beta T(u) = \beta(\alpha T)(u).$$

Similarly, if $T_1, T_2 \colon V \to W$ are linear, define their sum by

$$(T_1+T_2)(v) = T_1(v)+T_2(v).$$

The map T_1+T_2 is linear because

$$\begin{aligned}(T_1+T_2)(u+v) &= T_1(u+v)+T_2(u+v)\\ &= (T_1+T_2)(u)+(T_1+T_2)(v)\end{aligned}$$

and

$$(T_1+T_2)(\alpha u) = \alpha(T_1+T_2)(u).$$

Definition 13 (Space of linear maps)**.** *The collection of all linear transformations from V to W is denoted by*

$$\mathcal{L}(V,W) = \{T \colon V \to W : T \text{ is linear}\}.$$

Proposition 9. *The set $\mathcal{L}(V,W)$, equipped with pointwise addition and scalar multiplication, is a vector space over $\mathbb{F}$.*

Proof. The preceding calculations show that the two operations are closed in $\mathcal{L}(V,W)$. All vector-space identities follow pointwise from the corresponding identities in W. The additive identity is the zero transformation $0 \colon V \to W$, defined by $0(v) = 0$, and the additive inverse of T is the map $-T$, defined by $(-T)(v) = -T(v)$. □

6.4. **Composition of linear maps.** Let V, W, and Z be vector spaces over $\mathbb{F}$. Suppose that

$$T_2 \colon V \longrightarrow W \qquad \text{and} \qquad T_1 \colon W \longrightarrow Z$$

are linear. Their *composition* is the map

$$T_1 \circ T_2 \colon V \longrightarrow Z, \qquad (T_1 \circ T_2)(v) = T_1\big(T_2(v)\big).$$

Proposition 10. *The composition $T_1 \circ T_2$ is linear.*

Proof. For $u, v \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{aligned}(T_1 \circ T_2)(u+v) &= T_1\big(T_2(u) + T_2(v)\big) \\ &= T_1(T_2(u)) + T_1(T_2(v)),\end{aligned}$$

and

$$(T_1 \circ T_2)(\alpha u) = T_1\big(\alpha T_2(u)\big) = \alpha T_1(T_2(u)).$$

□

7. September 9, 2026

7.1. **Matrix operations and linear maps.** Let V and W be vector spaces over $\mathbb{F}$ with ordered bases

$$\mathcal{B} = (v_1, \ldots, v_n) \qquad \text{and} \qquad \mathcal{C} = (w_1, \ldots, w_m).$$

Suppose that $T, S \in \mathcal{L}(V, W)$ are represented by the matrices $A = [A_{ij}]$ and $B = [B_{ij}]$, respectively. Thus

$$T(v_j) = \sum_{i=1}^{m} A_{ij} w_i \qquad \text{and} \qquad S(v_j) = \sum_{i=1}^{m} B_{ij} w_i.$$

For each basis vector v_j,

$$\begin{aligned}(T+S)(v_j) &= T(v_j) + S(v_j) \\ &= \sum_{i=1}^{m} A_{ij} w_i + \sum_{i=1}^{m} B_{ij} w_i \\ &= \sum_{i=1}^{m} (A_{ij} + B_{ij}) w_i.\end{aligned}$$

Consequently, the matrix representing $T+S$ is obtained by adding the matrices representing T and S entrywise.

Definition 14 (Matrix addition)**.** *If $A, B \in M_{m\times n}(\mathbb{F})$, their sum is the matrix $A + B \in M_{m\times n}(\mathbb{F})$ defined by*

$$(A + B)_{ij} = A_{ij} + B_{ij}.$$

Similarly, if $\alpha \in \mathbb{F}$, then

$$(\alpha T)(v_j) = \alpha T(v_j) = \sum_{i=1}^{m} \alpha A_{ij} w_i.$$

Thus scalar multiplication of linear maps corresponds to entrywise scalar multiplication of matrices.

Definition 15 (Scalar multiplication of matrices)**.** *If $A \in M_{m\times n}(\mathbb{F})$ and $\alpha \in \mathbb{F}$, then $\alpha A \in M_{m\times n}(\mathbb{F})$ is defined by*

$$(\alpha A)_{ij} = \alpha A_{ij}.$$

7.2. **Composition and matrix multiplication.** Let Z be a third vector space with ordered basis

$$\mathcal{D} = (z_1, \ldots, z_p),$$

and consider linear maps

$$T_2 \colon V \longrightarrow W \qquad \text{and} \qquad T_1 \colon W \longrightarrow Z.$$

Suppose T_2 is represented by $B = [B_{kj}] \in M_{m\times n}(\mathbb{F})$ and T_1 is represented by $A = [A_{ik}] \in M_{p\times m}(\mathbb{F})$. For $j = 1, \ldots, n$,

$$\begin{aligned}
(T_1 \circ T_2)(v_j) &= T_1 \left(\sum_{k=1}^{m} B_{kj} w_k \right) \\
&= \sum_{k=1}^{m} B_{kj} T_1(w_k) \\
&= \sum_{k=1}^{m} B_{kj} \sum_{i=1}^{p} A_{ik} z_i \\
&= \sum_{i=1}^{p} \left(\sum_{k=1}^{m} A_{ik} B_{kj} \right) z_i.
\end{aligned}$$

The coefficient of z_i in this expression is the (i, j) entry of the matrix representing $T_1 \circ T_2$.

Definition 16 (Matrix multiplication)**.** *If $A \in M_{p\times m}(\mathbb{F})$ and $B \in M_{m\times n}(\mathbb{F})$, their product is the matrix $AB \in M_{p\times n}(\mathbb{F})$ whose entries are*

$$(AB)_{ij} = \sum_{k=1}^{m} A_{ik}B_{kj}.$$

Thus composition of linear maps corresponds to multiplication of their representing matrices, in the same order:

$$[T_1 \circ T_2] = [T_1][T_2].$$

Whenever all the products and sums involved are defined, matrix multiplication satisfies

$$\begin{aligned} (AB)C &= A(BC), \\ A(B+C) &= AB + AC, \\ (A+B)C &= AC + BC, \\ A(\alpha B) &= \alpha(AB) = (\alpha A)B. \end{aligned}$$

In general, however, $AB \neq BA$; the two products may even have different sizes, or one may be undefined. In particular, $M_{n\times n}(\mathbb{F})$, with matrix addition, scalar multiplication, and matrix multiplication, is an associative algebra over $\mathbb{F}$.

7.3. **Vector subspaces.**

Definition 17 (Subspace)**.** *Let V be a vector space over $\mathbb{F}$. A subset $U \subseteq V$ is a* subspace *of V if U, with the addition and scalar multiplication inherited from V, is itself a vector space over $\mathbb{F}$.*

Checking all the vector-space axioms is unnecessary. The following criterion reduces the verification to three conditions.

Proposition 11 (Subspace criterion)**.** *A subset $U \subseteq V$ is a subspace of V if and only if*

(i) $0 \in U$*;*
(ii) $u + v \in U$ *whenever* $u, v \in U$*;*
(iii) $\alpha u \in U$ *whenever* $u \in U$ *and* $\alpha \in \mathbb{F}$*.*

Equivalently, if U is known to be nonempty, it is enough to check the two closure conditions (ii) and (iii).

Example 15. *The affine line*

$$H = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : x = 2 \right\}$$

is not a subspace of $\mathbb{R}^2$, because it does not contain the zero vector.

The subspaces of $\mathbb{R}^2$ are precisely the zero subspace $\{0\}$, the lines through the origin, and $\mathbb{R}^2$ itself.

8. September 11, 2026

8.1. **The subspace criterion.**

Proof of Proposition 11. If U is a subspace of V, then U is a vector space under the operations inherited from V. It therefore contains the additive identity and is closed under addition and scalar multiplication.

Conversely, suppose conditions (i)–(iii) hold. The additive identity belongs to U by (i). If $u \in U$, then condition (iii), applied with the scalar -1, gives

$$-u = (-1)u \in U.$$

Hence U contains the additive inverse of each of its elements. The remaining vector-space axioms are identities already valid in V, so they remain valid when the vectors are restricted to U. Thus U is a subspace of V.

Finally, if U is nonempty and satisfies the two closure conditions, choose $u \in U$. Closure under scalar multiplication gives $0 = 0u \in U$, so condition (i) follows automatically. □

8.2. **Kernels and images.**

Definition 18 (Kernel and image)**.** *Let $T \in \mathcal{L}(V, W)$. The* kernel *and* image *of T are*

$$\ker T = \{v \in V : T(v) = 0\}$$

and

$$\operatorname{im} T = \{T(v) : v \in V\},$$

respectively. The image is also called the range *of T.*

Proposition 12. *If $T \in \mathcal{L}(V, W)$, then* $\ker T$ *is a subspace of V and* $\operatorname{im} T$ *is a subspace of W.*

Proof. Since $T(0) = 0$, both sets contain the appropriate zero vector. If $u, v \in \ker T$ and $\alpha \in \mathbb{F}$, then

$$T(u+v) = T(u) + T(v) = 0 \qquad \text{and} \qquad T(\alpha u) = \alpha T(u) = 0.$$

Thus $\ker T$ is a subspace. If $y_1 = T(v_1)$ and $y_2 = T(v_2)$ are in $\operatorname{im} T$, then

$$y_1 + y_2 = T(v_1 + v_2) \qquad \text{and} \qquad \alpha y_1 = T(\alpha v_1),$$

so the image is also a subspace. □

Example 16. *The continuous real-valued functions on $[0,1]$ form a subspace*

$$C([0,1]) \subseteq \{f : [0,1] \to \mathbb{R}\}$$

of the vector space of all real-valued functions on $[0,1]$.

Example 17. *Fix $b \in \mathbb{F}$ and let*

$$U_b = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{F}^4 : x_3 = 5x_4 + b \right\}.$$

Then U_b is a subspace of $\mathbb{F}^4$ if and only if $b = 0$. Indeed, the zero vector belongs to U_b only when $b = 0$, and when $b = 0$ the defining equation $x_3 = 5x_4$ is preserved by addition and scalar multiplication.

8.3. **Sums of subspaces.**

Definition 19 (Sum of subspaces)**.** *If $V_1, \ldots, V_m$ are subspaces of V, their* sum *is*

$$V_1 + \cdots + V_m = \{v_1 + \cdots + v_m : v_i \in V_i \text{ for } i = 1, \ldots, m\}.$$

Example 18. *Let*

$$U = \left\{ \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} : x \in \mathbb{F} \right\}, \qquad W = \left\{ \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix} : y \in \mathbb{F} \right\}.$$

Then

$$U + W = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{F} \right\}.$$

Proposition 13. *If $V_1, \ldots, V_m$ are subspaces of V, then $V_1 + \cdots + V_m$ is the smallest subspace of V containing each V_i.*

Proof. The sum contains $0 = 0 + \cdots + 0$. Adding two elements of the sum, or multiplying one by a scalar, can be done componentwise inside each V_i, so the sum is a subspace. For each i, any $v_i \in V_i$ can be written as

$$v_i = 0 + \cdots + 0 + v_i + 0 + \cdots + 0,$$

and hence $V_i \subseteq V_1 + \cdots + V_m$.

Conversely, let W be any subspace of V containing all the V_i. If $v = v_1 + \cdots + v_m$ with $v_i \in V_i$, then every v_i belongs to W. Since W is closed under addition, $v \in W$. Therefore $V_1 + \cdots + V_m \subseteq W$. □

8.4. **Direct sums.**

Definition 20 (Direct sum)**.** *Let $V_1, \ldots, V_m$ be subspaces of V. Their sum is a* direct sum *if every element of $V_1 + \cdots + V_m$ has a unique representation*

$$v = v_1 + \cdots + v_m, \qquad v_i \in V_i.$$

In this case we write

$$V_1 \oplus \cdots \oplus V_m.$$

Proposition 14 (Zero-sum criterion)**.** *The sum $V_1 + \cdots + V_m$ is direct if and only if the equation*

$$0 = v_1 + \cdots + v_m, \qquad v_i \in V_i,$$

implies $v_1 = \cdots = v_m = 0$.

Proof. If the sum is direct, then $0 = 0 + \cdots + 0$ is one representation of the zero vector, so uniqueness forces every other representation to have all components equal to zero.

Conversely, suppose the zero vector has only the trivial representation. If

$$v_1 + \cdots + v_m = u_1 + \cdots + u_m,$$

where $v_i, u_i \in V_i$, then

$$0 = (v_1 - u_1) + \cdots + (v_m - u_m).$$

The hypothesis gives $v_i - u_i = 0$ for each i, so $v_i = u_i$. Thus every representation is unique. □

Proposition 15. *If U and W are subspaces of V, then*

$$U + W = U \oplus W \quad \Longleftrightarrow \quad U \cap W = \{0\}.$$

Proof. Suppose first that $U + W$ is a direct sum, and let $v \in U \cap W$. Then

$$0 = v + (-v) = 0 + 0$$

gives two representations with one term in U and one term in W. Uniqueness implies $v = 0$.

Conversely, assume $U \cap W = \{0\}$. If

$$u + w = u' + w', \qquad u, u' \in U, \quad w, w' \in W,$$

then

$$u - u' = w' - w \in U \cap W.$$

Hence $u = u'$ and $w = w'$, so the representation is unique. □

For the coordinate-axis subspaces in the preceding example, $U \cap W = \{0\}$, and hence their sum is $U \oplus W$.

8.5. **Product vector spaces.**

Definition 21 (Product vector space)**.** *Let $V_1, \ldots, V_m$ be vector spaces over $\mathbb{F}$. Their Cartesian product is*

$$V_1 \times \cdots \times V_m = \{(v_1, \ldots, v_m) : v_i \in V_i\}.$$

Addition and scalar multiplication are defined coordinatewise:

$$(v_1, \dots, v_m) + (u_1, \dots, u_m) = (v_1 + u_1, \dots, v_m + u_m),$$
$$\alpha(v_1, \dots, v_m) = (\alpha v_1, \dots, \alpha v_m).$$

With these operations, $V_1 \times \cdots \times V_m$ is a vector space over $\mathbb{F}$; all vector-space axioms hold coordinatewise.

Example 19. *In* $\mathcal{P}_5(\mathbb{R}) \times \mathbb{R}^3$,

$$\left(5 - 6t + 4t^2, \begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix}\right) + \left(t + 9t^5, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}\right)$$
$$= \left(5 - 5t + 4t^2 + 9t^5, \begin{bmatrix} 5 \\ 10 \\ 9 \end{bmatrix}\right).$$

9. September 14, 2026

9.1. **Direct sums and injectivity.** Let $V_1, \dots, V_m$ be subspaces of a vector space V. Define the summation map

$$\Sigma \colon V_1 \times \cdots \times V_m \longrightarrow V_1 + \cdots + V_m$$

by

$$\Sigma(v_1, \dots, v_m) = v_1 + \cdots + v_m.$$

This map is linear and surjective.

Proposition 16. *The sum $V_1 + \cdots + V_m$ is direct if and only if the summation map Σ is injective.*

Proof. The map Σ is injective precisely when

$$\Sigma(v_1, \dots, v_m) = \Sigma(u_1, \dots, u_m)$$

implies

$$(v_1, \dots, v_m) = (u_1, \dots, u_m).$$

Equivalently, every vector in $V_1 + \cdots + V_m$ has only one representation as a sum of vectors from the respective subspaces. This is exactly the definition of a direct sum. □

9.2. **Quotient vector spaces.** Let U be a subset of V. For $v \in V$, define

$$v + U = \{v + u : u \in U\}.$$

The set $v + U$ is called a *translate* of U. When U is a subspace, a translate is also called a *coset* of U.

Example 20. *Let*

$$U = \left\{ \begin{bmatrix} x \\ 2x \end{bmatrix} : x \in \mathbb{R} \right\} \subseteq \mathbb{R}^2.$$

Then

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + U = \left\{ \begin{bmatrix} 2+x \\ 1+2x \end{bmatrix} : x \in \mathbb{R} \right\}.$$

The original subspace is the line $y = 2x$, while this translate is the parallel line $y = 2x - 3$.

Definition 22 (Quotient space)**.** *If U is a subspace of V, the* quotient space *of V by U is the set of all cosets of U:*

$$V/U = \{v + U : v \in V\}.$$

For the subspace $U \subseteq \mathbb{R}^2$ in the preceding example, $\mathbb{R}^2/U$ is the collection of all lines in $\mathbb{R}^2$ with slope 2.

Proposition 17. *Let U be a subspace of V and let $v, w \in V$. The following are equivalent:*

(i) $v - w \in U$;
(ii) $v + U = w + U$;
(iii) $(v + U) \cap (w + U) \neq \varnothing$.

Proof. Suppose first that $v - w \in U$. If $v + u \in v + U$, then

$$v + u = w + (v - w + u) \in w + U,$$

because $v - w + u \in U$. Thus $v + U \subseteq w + U$. Since $w - v = -(v - w) \in U$, the reverse inclusion follows in the same way, so $v + U = w + U$.

If $v + U = w + U$, then the two cosets plainly have nonempty intersection.

Finally, suppose $(v+U)\cap(w+U)\neq\varnothing$. There are $u_1,u_2\in U$ such that

$$v+u_1=w+u_2.$$

Therefore

$$v-w=u_2-u_1\in U.$$

□

Thus two cosets of a subspace are either equal or disjoint. We define addition and scalar multiplication on V/U by

$$(v+U)+(w+U)=(v+w)+U,$$
$$\alpha(v+U)=\alpha v+U.$$

Proposition 17 shows that these operations do not depend on the choice of representatives. With these operations, V/U is a vector space over $\mathbb{F}$, whose zero vector is the coset $U=0+U$.

9.3. **Finite-dimensional vector spaces.**

Definition 23 (Finite-dimensional vector space)**.** *A vector space V is* finite-dimensional *if some finite collection of vectors spans V. It is* infinite-dimensional *if it is not finite-dimensional.*

Example 21. *The coordinate space $\mathbb{F}^n$ is finite-dimensional because*

$$\mathbb{F}^n=\operatorname{span}\{e_1,\dots,e_n\}.$$

Likewise, the space of polynomials of degree at most m is finite-dimensional because

$$\mathcal{P}_m(\mathbb{F})=\operatorname{span}\{1,t,t^2,\dots,t^m\}.$$

Example 22. *The vector space $\mathcal{P}(\mathbb{F})$ of all polynomials with coefficients in $\mathbb{F}$ is infinite-dimensional. Indeed, suppose a finite collection of polynomials spanned $\mathcal{P}(\mathbb{F})$, and let m be the largest degree among the nonzero polynomials in the collection. Every polynomial in their span would have degree at most m, so t^{m+1} could not belong to the span, a contradiction.*

10. September 16, 2026

10.1. **Comparing independent and spanning collections.**

Proposition 18. *Suppose that $u_1, \dots, u_m$ is a linearly independent collection in V and that $w_1, \dots, w_n$ spans V. Then*

$$m \leq n.$$

Thus, in a finite-dimensional vector space, every linearly independent collection has cardinality at most that of every finite spanning collection.

Proof. We successively insert the vectors $u_1, u_2, \dots$ into the spanning list $w_1, \dots, w_n$, removing one of the w's after each insertion.

First adjoin u_1 to obtain

$$u_1, w_1, \dots, w_n.$$

This collection is linearly dependent because $w_1, \dots, w_n$ spans V. Since $u_1 \neq 0$, one of the w's is a linear combination of the other vectors. Remove that vector. The resulting collection still spans V and has n vectors.

Inductively, suppose that after $k-1$ steps we have a spanning collection of n vectors consisting of $u_1, \dots, u_{k-1}$ and $n-k+1$ of the original w's. Adjoin u_k. The resulting collection is dependent. In a nontrivial dependence relation, some remaining w must have a nonzero coefficient; otherwise the relation would contradict the linear independence of $u_1, \dots, u_k$. We can therefore remove that w without changing the span.

If $m > n$, then after n steps the collection $u_1, \dots, u_n$ would span V. In particular, u_{n+1} would lie in their span, contradicting the linear independence of $u_1, \dots, u_m$. Hence $m \leq n$. □

Example 23. *No collection of four vectors in $\mathbb{R}^3$ is linearly independent, because the three standard basis vectors span $\mathbb{R}^3$. Similarly, no collection of three vectors can span $\mathbb{R}^4$, because the four standard basis vectors are linearly independent in $\mathbb{R}^4$.*

10.2. **Subspaces of finite-dimensional spaces.**

Proposition 19. *Every subspace of a finite-dimensional vector space is finite-dimensional.*

Proof. Let U be a subspace of a finite-dimensional vector space V. If $U = \{0\}$, then the empty collection spans U, so U is finite-dimensional.

Otherwise choose a nonzero vector $u_1 \in U$. If $U = \operatorname{span}\{u_1\}$, we are done. If not, choose

$$u_2 \in U \setminus \operatorname{span}\{u_1\}.$$

Continue in this fashion: once $u_1, \ldots, u_{k-1}$ have been chosen, stop if they span U; otherwise choose

$$u_k \in U \setminus \operatorname{span}\{u_1, \ldots, u_{k-1}\}.$$

At every stage the chosen collection is linearly independent. Since V has a finite spanning collection, Proposition 18 bounds the number of vectors in any linearly independent collection in V. The process must therefore stop after finitely many steps, at which point the chosen vectors span U. □

Corollary 1. *Every finite-dimensional vector space has a basis.*

Proof. By definition, a finite-dimensional vector space has a finite spanning collection. Removing redundant vectors from that collection produces a basis. □

10.3. **Extending a linearly independent collection.**

Proposition 20 (Basis extension theorem)**.** *Every finite linearly independent collection in a finite-dimensional vector space can be extended to a basis of the space.*

Proof. Let $u_1, \ldots, u_m$ be linearly independent in V, and choose a finite spanning collection $w_1, \ldots, w_n$ for V. The combined collection

$$u_1, \ldots, u_m, w_1, \ldots, w_n$$

spans V. Examine the w's in order. Whenever a vector w_j lies in the span of the vectors preceding it, remove it. Removing such a vector

does not change the span, and none of the u_i is removed. After all the w's have been considered, the remaining collection spans V and is linearly independent. It is therefore a basis containing $u_1, \ldots, u_m$. $\square$

Example 24. *Consider the linearly independent vectors*

$$v_1 = \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} -9 \\ 6 \\ 8 \end{bmatrix}$$

in $\mathbb{R}^3$*. Adjoin the standard basis vectors* e_1, e_2, e_3*. Since*

$$e_1 = -\frac{1}{5}(v_2 - 2v_1),$$

the vector e_1 *is redundant. On the other hand,* e_2 *is not in* $\operatorname{span}\{v_1, v_2\}$*, so we retain it. Finally,*

$$e_3 = -\frac{1}{10}\left(v_2 - \frac{9}{2}v_1 + \frac{15}{2}e_2\right),$$

so e_3 *is redundant. Therefore*

$$\left\{ \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -9 \\ 6 \\ 8 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

is a basis of $\mathbb{R}^3$ *extending the original collection.*

11. September 18, 2026

11.1. Complements of subspaces.

Proposition 21. *Suppose that* V *is finite-dimensional and that* U *is a subspace of* V*. Then there is a subspace* W *of* V *such that*

$$V = U \oplus W.$$

Proof. Because V is finite-dimensional, so is U. Let $u_1, \ldots, u_m$ be a basis of U. This collection is linearly independent in V, so the basis extension theorem gives a basis

$$u_1, \ldots, u_m, w_1, \ldots, w_n$$

of V. Let

$$W = \operatorname{span}\{w_1, \ldots, w_n\}.$$

Certainly $U + W \subseteq V$. Conversely, if $v \in V$, then there are scalars $a_1, \ldots, a_m, b_1, \ldots, b_n$ such that

$$v = \underbrace{a_1u_1 + \cdots + a_mu_m}_{\in U} + \underbrace{b_1w_1 + \cdots + b_nw_n}_{\in W}.$$

Hence $v \in U + W$, and therefore $V = U + W$.

It remains to show that $U \cap W = \{0\}$. Let $v \in U \cap W$. Then

$$v = a_1u_1 + \cdots + a_mu_m = b_1w_1 + \cdots + b_nw_n$$

for some scalars a_i and b_j. Thus

$$a_1u_1 + \cdots + a_mu_m - b_1w_1 - \cdots - b_nw_n = 0.$$

Because the combined collection is a basis of V, all the coefficients are zero. In particular, $v = 0$, so $U \cap W = \{0\}$. Thus $V = U \oplus W$. □

11.2. **Dimension.**

Proposition 22. *Any two bases of a finite-dimensional vector space have the same cardinality.*

Proof. Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases of a finite-dimensional vector space V. Because $\mathcal{B}_1$ is linearly independent and $\mathcal{B}_2$ spans V, Proposition 18 gives

$$|\mathcal{B}_1| \leq |\mathcal{B}_2|.$$

Interchanging the roles of the two bases gives

$$|\mathcal{B}_2| \leq |\mathcal{B}_1|.$$

Therefore $|\mathcal{B}_1| = |\mathcal{B}_2|$. □

Definition 24 (Dimension)**.** *The* dimension *of a finite-dimensional vector space V is the cardinality of any basis of V. It is denoted by* $\dim V$.

Example 25. *The standard basis* $\{e_1, \ldots, e_n\}$ *of* $\mathbb{F}^n$ *has* n *elements, so*

$$\dim \mathbb{F}^n = n.$$

Example 26. *The standard basis* $\{1, x, \ldots, x^m\}$ *of* $\mathcal{P}_m(\mathbb{F})$ *has* $m+1$ *elements, so*

$$\dim \mathcal{P}_m(\mathbb{F}) = m + 1.$$

Example 27. *Let*

$$U = \left\{ \begin{bmatrix} x \\ 0 \\ y \end{bmatrix} : x, y \in \mathbb{F} \right\}.$$

Because

$$U = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

and the two displayed vectors are linearly independent, they form a basis of U*. Hence* $\dim U = 2$*.*

Example 28. *As a vector space over* $\mathbb{R}$*, the space* $\mathbb{R}^2$ *has dimension* 2*. As a vector space over* $\mathbb{C}$*, the space* $\mathbb{C}$ *has dimension* 1*. Although* $\mathbb{R}^2$ *and* $\mathbb{C}$ *are often identified as sets, the choice of scalar field is essential when determining dimension.*

11.3. **Consequences of dimension.**

Proposition 23. *If* V *is finite-dimensional and* U *is a subspace of* V*, then*

$$\dim U \leq \dim V.$$

Proof. A basis of U is linearly independent in V, while a basis of V spans V. The result follows from Proposition 18. □

Proposition 24. *Suppose that* V *is finite-dimensional. Every linearly independent collection of* $\dim V$ *vectors in* V *is a basis of* V*.*

Proof. Write $\dim V = n$, and suppose that $v_1, \ldots, v_n$ is linearly independent in V. By the basis extension theorem, this collection can be

extended to a basis of V. Every basis of V has exactly n vectors, however, so no vectors need to be adjoined. Thus $v_1, \ldots, v_n$ is already a basis of V. □

Corollary 2. *Suppose that V is finite-dimensional and that U is a subspace of V. If $\dim U = \dim V$, then $U = V$.*

Proof. Let $u_1, \ldots, u_n$ be a basis of U. Then $n = \dim U = \dim V$, and $u_1, \ldots, u_n$ is linearly independent in V. By Proposition 24, it is a basis of V. Hence every vector in V is a linear combination of vectors in U, so $V \subseteq U$. The reverse inclusion holds because U is a subspace of V. Therefore $U = V$. □

Example 29. *Let*

$$U = \{p \in \mathcal{P}_3(\mathbb{R}) : p'(5) = 0\}.$$

We find a basis of U. Write

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3.$$

Because

$$p'(x) = a_1 + 2a_2 x + 3a_3 x^2,$$

the condition $p'(5) = 0$ is equivalent to

$$a_1 + 10a_2 + 75a_3 = 0, \qquad \text{or equivalently} \qquad a_1 = -10a_2 - 75a_3.$$

Therefore every $p \in U$ has the form

$$\begin{aligned} p(x) &= a_0 + (-10a_2 - 75a_3)x + a_2 x^2 + a_3 x^3 \\ &= a_0 + a_2(x^2 - 10x) + a_3(x^3 - 75x). \end{aligned}$$

Thus the collection

$$\{1, x^2 - 10x, x^3 - 75x\}$$

spans U. It is linearly independent because its members have distinct degrees. Consequently,

$$\boxed{\{1, x^2 - 10x, x^3 - 75x\}}$$

is a basis of U, and $\dim U = 3$.

12. September 21, 2026

12.1. **Spanning collections and dimension.**

Proposition 25. *Suppose that V is finite-dimensional. Every collection of $\dim V$ vectors that spans V is a basis of V.*

Proof. Write $\dim V = n$, and suppose that $v_1, \ldots, v_n$ spans V. This collection can be reduced to a basis by removing redundant vectors. However, every basis of V has exactly n vectors, so no vectors can be removed. Thus $v_1, \ldots, v_n$ is already a basis of V. □

12.2. **Dimension of a sum of subspaces.**

Proposition 26. *If V_1 and V_2 are subspaces of a finite-dimensional vector space, then*

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

Proof. Let $v_1, \ldots, v_m$ be a basis of $V_1 \cap V_2$, so that $\dim(V_1 \cap V_2) = m$. This collection is linearly independent in V_1, so the basis extension theorem gives a basis

$$v_1, \ldots, v_m, u_1, \ldots, u_j$$

of V_1. Similarly, extend $v_1, \ldots, v_m$ to a basis

$$v_1, \ldots, v_m, w_1, \ldots, w_k$$

of V_2. Hence

$$\dim V_1 = m + j \qquad \text{and} \qquad \dim V_2 = m + k.$$

We will show that

$$v_1, \ldots, v_m, u_1, \ldots, u_j, w_1, \ldots, w_k$$

is a basis of $V_1 + V_2$.

All the displayed vectors belong to $V_1 + V_2$, so their span is contained in $V_1 + V_2$. Conversely, every vector in $V_1 + V_2$ is a sum of a vector in V_1 and a vector in V_2. Expressing each summand in the corresponding basis shows that the combined collection spans $V_1 + V_2$.

To prove linear independence, suppose that

$$a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j + c_1w_1 + \cdots + c_kw_k = 0. \tag{1}$$

Rearranging gives

$$c_1w_1 + \cdots + c_kw_k = -a_1v_1 - \cdots - a_mv_m - b_1u_1 - \cdots - b_ju_j.$$

The right-hand side belongs to V_1, while the left-hand side belongs to V_2. Therefore

$$c_1w_1 + \cdots + c_kw_k \in V_1 \cap V_2.$$

Since $v_1, \ldots, v_m$ is a basis of the intersection, there are scalars $d_1, \ldots, d_m$ such that

$$c_1w_1 + \cdots + c_kw_k = d_1v_1 + \cdots + d_mv_m.$$

Thus

$$c_1w_1 + \cdots + c_kw_k - d_1v_1 - \cdots - d_mv_m = 0.$$

The collection $v_1, \ldots, v_m, w_1, \ldots, w_k$ is a basis of V_2 and hence linearly independent. Consequently, all the c's and d's are zero. Equation (1) now becomes

$$a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j = 0.$$

Because $v_1, \ldots, v_m, u_1, \ldots, u_j$ is a basis of V_1, all the a's and b's are also zero. This proves linear independence.

The combined collection is therefore a basis of $V_1 + V_2$, and

$$\begin{aligned}\dim(V_1 + V_2) &= m + j + k \\ &= (m + j) + (m + k) - m \\ &= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).\end{aligned}$$

□

12.3. **Linear maps on finite-dimensional spaces.**

Theorem 2 (Fundamental theorem of linear maps)**.** *Suppose that V is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Then $\operatorname{im} T$ is finite-dimensional and*

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

Recall that

$$\ker T = \{v \in V : T(v) = 0\} \qquad \text{and} \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

Before proving the theorem, we connect the kernel and image with injectivity and surjectivity, beginning with injectivity.

Proposition 27. *Let $T \in \mathcal{L}(V, W)$. Then T is injective if and only if $\ker T = \{0\}$.*

Proof. First note that every linear map sends zero to zero, since

$$T(0) = T(0 + 0) = T(0) + T(0)$$

implies $T(0) = 0$. This does not require injectivity. In particular, $\{0\} \subseteq \ker T$.

Suppose that T is injective. If $v \in \ker T$, then

$$T(v) = 0 = T(0).$$

Injectivity gives $v = 0$, so $\ker T \subseteq \{0\}$. Therefore $\ker T = \{0\}$.

Conversely, suppose that $\ker T = \{0\}$. Let $v_1, v_2 \in V$ satisfy $T(v_1) = T(v_2)$. By linearity,

$$T(v_1 - v_2) = T(v_1) - T(v_2) = 0.$$

Hence $v_1 - v_2 \in \ker T = \{0\}$, so $v_1 = v_2$. Thus T is injective. □

13. September 23, 2026

13.1. **Injectivity, surjectivity, and dimension.** Recall from Proposition 27 that a linear map $T \in \mathcal{L}(V, W)$ is injective if and only if $\ker T = \{0\}$. In particular, if $\ker T = \{0\}$, then

$$T(v) = T(v') \implies T(v - v') = 0 \implies v - v' = 0 \implies v = v'.$$

The corresponding criterion for surjectivity follows directly from the definition of the image:

$$T \text{ is surjective} \iff \operatorname{im} T = W.$$

13.2. **Proof of the fundamental theorem of linear maps.** We now prove Theorem 2: if V is finite-dimensional and $T \in \mathcal{L}(V, W)$,

then $\operatorname{im} T$ is finite-dimensional and

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

This identity is also called the *rank–nullity formula*; $\dim \operatorname{im} T$ is the rank and $\dim \ker T$ is the nullity.

Proof of Theorem 2. Since $\ker T$ is a subspace of the finite-dimensional space V, it has a basis $u_1, \ldots, u_m$. Extend this collection to a basis

$$u_1, \ldots, u_m, v_1, \ldots, v_n$$

of V. Thus $\dim \ker T = m$ and $\dim V = m + n$. We claim that

$$T(v_1), \ldots, T(v_n)$$

is a basis of $\operatorname{im} T$.

First, every $v \in V$ can be written as

$$v = a_1 u_1 + \cdots + a_m u_m + b_1 v_1 + \cdots + b_n v_n.$$

Because each u_i belongs to $\ker T$, linearity gives

$$T(v) = b_1 T(v_1) + \cdots + b_n T(v_n).$$

Hence $\operatorname{im} T$ is contained in the span of the displayed images. The reverse inclusion holds because each $T(v_j)$ belongs to the subspace $\operatorname{im} T$. Therefore

$$\operatorname{im} T = \operatorname{span}\{T(v_1), \ldots, T(v_n)\}.$$

To show linear independence, suppose that

$$c_1 T(v_1) + \cdots + c_n T(v_n) = 0.$$

Then $T(c_1 v_1 + \cdots + c_n v_n) = 0$, so $c_1 v_1 + \cdots + c_n v_n \in \ker T$. There are scalars $d_1, \ldots, d_m$ such that

$$c_1 v_1 + \cdots + c_n v_n = d_1 u_1 + \cdots + d_m u_m.$$

Rearranging gives

$$-d_1 u_1 - \cdots - d_m u_m + c_1 v_1 + \cdots + c_n v_n = 0.$$

The chosen basis of V is linearly independent, so all the coefficients are zero. In particular, $c_1 = \cdots = c_n = 0$.

Thus the image has a basis of n vectors, so $\dim \operatorname{im} T = n$. It follows that

$$\dim V = m + n = \dim \ker T + \dim \operatorname{im} T.$$

The same argument includes $m = 0$ or $n = 0$, using the empty basis for the zero subspace. □

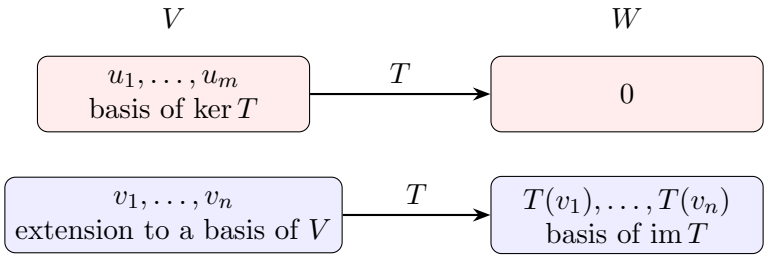


FIGURE 1. The kernel basis maps to zero; the images of the remaining basis vectors form a basis of the image.

13.3. **Dimension restrictions on linear maps.**

Corollary 3. *Let V and W be finite-dimensional vector spaces. If $\dim V > \dim W$, then no linear map from V to W is injective.*

Proof. If $T \in \mathcal{L}(V, W)$, then $\operatorname{im} T \subseteq W$, so $\dim \operatorname{im} T \leq \dim W$. Rank–nullity gives

$$\dim \ker T = \dim V - \dim \operatorname{im} T \geq \dim V - \dim W > 0.$$

Hence $\ker T \neq \{0\}$, and T is not injective. □

Example 30. *Define $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ by*

$$T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \sqrt{7}\, x_1 + \pi x_2 + x_4 \\ 97x_1 + 3x_2 + 2x_3 \\ x_2 + 6x_3 + 7x_4 \end{bmatrix}.$$

Since $4 > 3$, this map is not injective, regardless of the particular coefficients. There is no need to solve for its kernel explicitly. The same example works over $\mathbb{C}$.

Proposition 28. *Suppose that V and W are finite-dimensional and $\dim V < \dim W$. Then no linear map from V to W is surjective.*

Proof. For $T \in \mathcal{L}(V, W)$, rank–nullity gives

$$\dim \operatorname{im} T = \dim V - \dim \ker T \leq \dim V < \dim W.$$

Thus $\operatorname{im} T \neq W$, so T is not surjective. □

13.4. **A complement to the kernel.**

Proposition 29. *Let V be finite-dimensional, and let $T \in \mathcal{L}(V, W)$. There is a subspace $U \subseteq V$ such that*

$$V = \ker T \oplus U \qquad \text{and} \qquad \operatorname{im} T = \{T(u) : u \in U\}.$$

In particular, $U \cap \ker T = \{0\}$, and the restriction $T|_U \colon U \to \operatorname{im} T$ is bijective.

Proof. As in the proof of rank–nullity, extend a basis $u_1, \ldots, u_m$ of $\ker T$ to a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ of V. Set

$$U = \operatorname{span}\{v_1, \ldots, v_n\}.$$

The combined basis spans V, so $V = \ker T + U$. If $w \in U \cap \ker T$, then

$$w = b_1 v_1 + \cdots + b_n v_n = a_1 u_1 + \cdots + a_m u_m.$$

Linear independence of the combined basis forces every coefficient to vanish. Thus $w = 0$, and the sum is direct.

Every $v \in V$ has the form $v = k + u$, where $k \in \ker T$ and $u \in U$. Therefore $T(v) = T(u)$, proving that $T|_U$ maps onto $\operatorname{im} T$. Its kernel is $U \cap \ker T = \{0\}$, so it is also injective. □

14. September 25, 2026

14.1. **Invertible linear maps.** Write Id_V for the identity map on V, defined by $\mathrm{Id}_V(v) = v$. As with matrix multiplication, we sometimes write ST for the composition $S \circ T$.

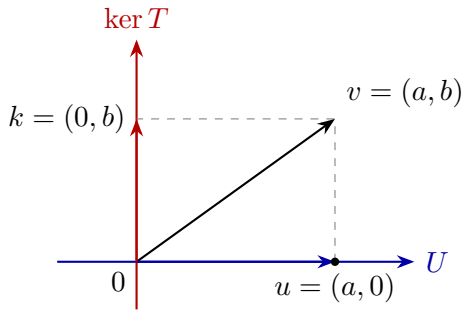


FIGURE 2. For $T\colon \mathbb{R}^2 \to \mathbb{R}$, $T(x, y) = x$, take U to be the horizontal axis. The decomposition $v = k + u$ gives $T(v) = T(u) = a$, while $T(k) = 0$.

Definition 25 (Invertibility)**.** *A linear map $T \in \mathcal{L}(V, W)$ is* invertible *if there exists a linear map $S \in \mathcal{L}(W, V)$ such that*

$$S \circ T = \mathrm{Id}_V \qquad \text{and} \qquad T \circ S = \mathrm{Id}_W .$$

Such a map S is called an inverse *of T.*

Proposition 30 (Uniqueness of the inverse)**.** *An invertible linear map has a unique inverse.*

Proof. Suppose $S_1, S_2 \in \mathcal{L}(W, V)$ are both inverses of T. By associativity of composition,

$$\begin{aligned} S_1 &= S_1 \circ \mathrm{Id}_W \\ &= S_1 \circ (T \circ S_2) \\ &= (S_1 \circ T) \circ S_2 \\ &= \mathrm{Id}_V \circ S_2 = S_2. \end{aligned}$$

□

We denote the unique inverse by $T^{-1} \in \mathcal{L}(W, V)$. Thus

$$T^{-1}T = \mathrm{Id}_V \qquad \text{and} \qquad TT^{-1} = \mathrm{Id}_W .$$

Example 31. *Define* $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ *by*

$$T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -y \\ x \\ 4z \end{bmatrix}.$$

Its values on the standard basis are

$$T(e_1) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad T(e_2) = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \qquad T(e_3) = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}.$$

Hence its matrix with respect to the standard basis is

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

To recover the input from an output (x, y, z)*, reverse the first two coordinates with the appropriate sign and divide the third coordinate by* 4*. This gives*

$$T^{-1}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ -x \\ z/4 \end{bmatrix}, \qquad A^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}.$$

Indeed, direct multiplication gives $AA^{-1} = A^{-1}A = I_3$*. Equivalently, augment* A *by the identity matrix and perform the row operations* $R_1 \leftrightarrow R_2$*,* $R_2 \leftarrow -R_2$*, and* $R_3 \leftarrow \frac{1}{4}R_3$*:*

$$\left[\begin{array}{ccc|ccc} 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 4 & 0 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{4} \end{array}\right].$$

14.2. **Invertibility and bijectivity.**

Proposition 31. *A linear map is invertible if and only if it is bijective.*

Proof. Let $T \in \mathcal{L}(V, W)$. Suppose first that T is invertible. If $T(u) = T(v)$, applying T^{-1} gives

$$u = T^{-1}(T(u)) = T^{-1}(T(v)) = v.$$

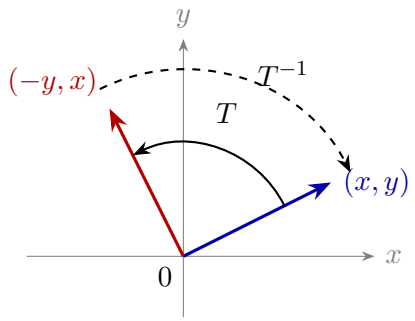


FIGURE 3. In the xy-plane, T rotates by 90° counterclockwise and T^{-1} rotates clockwise. In the third coordinate, T multiplies by 4 and T^{-1} divides by 4.

Thus T is injective. For every $w \in W$, the vector $v = T^{-1}(w)$ satisfies

$$T(v) = T(T^{-1}(w)) = w,$$

so T is also surjective.

Conversely, suppose that T is bijective. For each $w \in W$, surjectivity provides $v \in V$ with $T(v) = w$, and injectivity makes this vector unique. Define $S\colon W \to V$ by assigning to w this unique vector. By construction, $T \circ S = \mathrm{Id}_W$. Also, for $v \in V$,

$$T(S(T(v))) = T(v),$$

so injectivity gives $S(T(v)) = v$. Thus $S \circ T = \mathrm{Id}_V$.

It remains to check that S is linear. For $w_1, w_2 \in W$,

$$\begin{aligned} T(S(w_1) + S(w_2)) &= T(S(w_1)) + T(S(w_2)) \\ &= w_1 + w_2 \\ &= T(S(w_1 + w_2)). \end{aligned}$$

Injectivity of T yields

$$S(w_1 + w_2) = S(w_1) + S(w_2).$$

Similarly, for $\lambda \in \mathbb{F}$ and $w \in W$,

$$T(\lambda S(w)) = \lambda T(S(w)) = \lambda w = T(S(\lambda w)),$$

so $S(\lambda w) = \lambda S(w)$. Therefore $S \in \mathcal{L}(W, V)$ and S is an inverse of T. □

14.3. **Why the dimension hypotheses matter.** In general, injectivity alone or surjectivity alone does not imply invertibility.

Example 32. *On the space $\mathcal{P}(\mathbb{R})$ of all real polynomials, define*

$$M\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R}), \qquad (Mp)(x) = x^2 p(x).$$

This map is linear and injective: if $x^2 p(x)$ is the zero polynomial, then p is the zero polynomial. It is not surjective, because the constant polynomial 1 does not belong to its image. Thus M is not invertible, even though its domain and codomain are the same vector space.

Example 33 (A complementary example)**.** *Differentiation*

$$D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R}), \qquad D(p) = p',$$

is surjective, since every real polynomial has a polynomial antiderivative. It is not injective, since every constant polynomial belongs to $\ker D$. Therefore D is not invertible.

Proposition 32. *Suppose that V and W are finite-dimensional with $\dim V = \dim W$, and let $T \in \mathcal{L}(V, W)$. Then the following are equivalent:*

(i) T is invertible;
(ii) T is injective;
(iii) T is surjective.

Proof. By rank–nullity and the equality of dimensions,

$$\begin{aligned} T \text{ is injective} &\iff \ker T = \{0\} \\ &\iff \dim \ker T = 0 \\ &\iff \dim \operatorname{im} T = \dim V = \dim W \\ &\iff \operatorname{im} T = W \\ &\iff T \text{ is surjective.} \end{aligned}$$

Here we used the fact that a subspace of a finite-dimensional space with the same dimension is the whole space. Thus either injectivity or surjectivity implies bijectivity, which is equivalent to invertibility by Proposition 31. □

14.4. **One-sided inverses in equal dimensions.**

Proposition 33. *Suppose that V and W are finite-dimensional with $\dim V = \dim W$. If $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, V)$, then*

$$ST = \mathrm{Id}_V \quad \Longleftrightarrow \quad TS = \mathrm{Id}_W .$$

In either case, $S = T^{-1}$.

Proof. Suppose $ST = \mathrm{Id}_V$. If $v \in \ker T$, then

$$v = (ST)(v) = S(T(v)) = S(0) = 0.$$

Hence T is injective, and Proposition 32 makes it invertible. Consequently,

$$S = S(TT^{-1}) = (ST)T^{-1} = T^{-1},$$

which implies $TS = \mathrm{Id}_W$.

Conversely, suppose $TS = \mathrm{Id}_W$. For every $w \in W$,

$$w = T(S(w)),$$

so T is surjective and hence invertible by the same proposition. Therefore

$$S = (T^{-1}T)S = T^{-1}(TS) = T^{-1},$$

and $ST = \mathrm{Id}_V$. □

15. September 28, 2026

15.1. **Isomorphic vector spaces.**

Definition 26 (Isomorphism)**.** *An* isomorphism *is an invertible linear map. Two vector spaces V and W over the same field $\mathbb{F}$ are* isomorphic *if there exists an isomorphism from V onto W. We write $V \cong W$.*

By Proposition 31, a linear map is an isomorphism precisely when it is both injective and surjective.

Proposition 34. *Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.*

Proof. Suppose first that $T\colon V \to W$ is an isomorphism. Then T is injective and surjective, so $\ker T = \{0\}$ and $\operatorname{im} T = W$. By the fundamental theorem of linear maps (Theorem 2),

$$\dim V = \dim \ker T + \dim \operatorname{im} T = 0 + \dim W = \dim W.$$

Conversely, let $\dim V = \dim W = n$. Choose bases $v_1, \ldots, v_n$ of V and $w_1, \ldots, w_n$ of W. By the linear map lemma (Lemma 2), there is a unique linear map $T\colon V \to W$ satisfying $T(v_j) = w_j$ for all j. Explicitly,

$$T(c_1 v_1 + \cdots + c_n v_n) = c_1 w_1 + \cdots + c_n w_n.$$

This map is well-defined because coordinates with respect to a basis are unique. Since $w_1, \ldots, w_n$ spans W, every vector in W occurs as an output, so T is surjective. If $T(c_1 v_1 + \cdots + c_n v_n) = 0$, then

$$c_1 w_1 + \cdots + c_n w_n = 0.$$

Linear independence of the w_j gives $c_1 = \cdots = c_n = 0$. Thus $\ker T = \{0\}$, and Proposition 27 shows that T is injective. Therefore T is an isomorphism. When $n = 0$, the unique map between the two zero spaces is an isomorphism as well. □

15.2. **The space of linear maps and the space of matrices.** Fix ordered bases

$$\mathcal{B} = (v_1, \ldots, v_n) \qquad \text{and} \qquad \mathcal{C} = (w_1, \ldots, w_m)$$

of V and W, respectively. To keep track of both bases, write $[T]_{\mathcal{C}}^{\mathcal{B}}$ for the matrix of $T \in \mathcal{L}(V, W)$, characterized by

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C}}^{\mathcal{B}} [v]_{\mathcal{B}}.$$

Thus the jth column is $[T(v_j)]_{\mathcal{C}}$, as in Theorem 1. The superscript specifies the domain basis and the subscript specifies the codomain basis. Thus the matrix takes $\mathcal{B}$-coordinates to $\mathcal{C}$-coordinates.

Proposition 35. *The map*

$$\mathcal{M}\colon \mathcal{L}(V,W) \longrightarrow M_{m\times n}(\mathbb{F}), \qquad \mathcal{M}(T) = [T]_{\mathcal{C}}^{\mathcal{B}},$$

is an isomorphism of vector spaces.

Proof. The calculations in Section 7.1 show that

$$\mathcal{M}(T+S) = \mathcal{M}(T) + \mathcal{M}(S), \qquad \mathcal{M}(\alpha T) = \alpha \mathcal{M}(T).$$

Hence $\mathcal{M}$ is linear.

For injectivity, suppose that $\mathcal{M}(T)$ is the zero matrix. Every column is zero, so for each j,

$$T(v_j) = 0w_1 + \cdots + 0w_m = 0.$$

Since the v_j form a basis, linearity implies that $T(v) = 0$ for every $v \in V$. Thus T is the zero map, and $\ker \mathcal{M} = \{0\}$. By Proposition 27, $\mathcal{M}$ is injective.

For surjectivity, take an arbitrary matrix $A = [A_{ij}] \in M_{m\times n}(\mathbb{F})$. Prescribe

$$T(v_j) = \sum_{i=1}^{m} A_{ij} w_i \qquad (j = 1, \ldots, n).$$

Lemma 2 gives a linear map $T \in \mathcal{L}(V,W)$ with these values, and its matrix is exactly A. Therefore $\mathcal{M}$ is surjective. Being linear and bijective, it is an isomorphism by Proposition 31. □

Corollary 4. *If V and W are finite-dimensional, then $\mathcal{L}(V,W)$ is finite-dimensional and*

$$\dim \mathcal{L}(V,W) = (\dim V)(\dim W).$$

Proof. Set $n = \dim V$ and $m = \dim W$. For $1 \le i \le m$ and $1 \le j \le n$, let E_{ij} be the matrix with a 1 in entry (i,j) and zeros elsewhere. Every matrix $A = [A_{ij}]$ has the unique expansion

$$A = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} E_{ij}.$$

Thus the mn matrices E_{ij} form a basis of $M_{m\times n}(\mathbb{F})$. Because $\mathcal{M}$ is a linear isomorphism, the maps $\mathcal{M}^{-1}(E_{ij})$ form a basis of $\mathcal{L}(V,W)$.

Hence this space is finite-dimensional with dimension mn. If either m or n is zero, the space of linear maps is the zero space, and the same formula holds. □

15.3. **Invertible matrices.**

Definition 27 (Invertible matrix)**.** *A square matrix $A \in M_{n\times n}(\mathbb{F})$ is* invertible *if there exists $B \in M_{n\times n}(\mathbb{F})$ such that*

$$AB = BA = I_n,$$

where I_n is the identity matrix. We call B the inverse *of A and denote it by A^{-1}.*

The inverse is unique: if B and C are both inverses of A, then

$$B = B(AC) = (BA)C = C.$$

Let V and W be finite-dimensional, and let $T\colon V \to W$ be an isomorphism. By Proposition 34, both spaces have the same dimension n. Let

$$A = [T]_{\mathcal{C}}^{\mathcal{B}}, \qquad B = [T^{-1}]_{\mathcal{B}}^{\mathcal{C}}.$$

By Section 7.2, composition corresponds to matrix multiplication. Thus $T^{-1}T = \mathrm{Id}_V$ and $TT^{-1} = \mathrm{Id}_W$ give

$$BA = I_n \qquad \text{and} \qquad AB = I_n.$$

Consequently,

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = \left([T]_{\mathcal{C}}^{\mathcal{B}}\right)^{-1}.$$

Notice that the roles of the domain and codomain bases are reversed for the inverse map.

15.4. **Change of basis.** Let

$$\mathcal{B} = (v_1, \ldots, v_n) \qquad \text{and} \qquad \mathcal{C} = (u_1, \ldots, u_n)$$

be two ordered bases of the same vector space V. Define

$$P = [\mathrm{Id}_V]_{\mathcal{C}}^{\mathcal{B}}, \qquad Q = [\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{C}}.$$

Although the identity map leaves each vector unchanged, its matrix need not be I_n when the domain and codomain use different bases. Here

$$[v]_{\mathcal{C}} = P[v]_{\mathcal{B}}, \qquad [v]_{\mathcal{B}} = Q[v]_{\mathcal{C}}.$$

In particular, the columns of P are the old basis vectors expressed in the new basis:

$$P = \begin{bmatrix} | & & | \\ [v_1]_{\mathcal{C}} & \cdots & [v_n]_{\mathcal{C}} \\ | & & | \end{bmatrix}.$$

Proposition 36. *The matrices P and Q are invertible, and each is the inverse of the other.*

Proof. Changing from $\mathcal{B}$-coordinates to $\mathcal{C}$-coordinates and then back represents the composition $\mathrm{Id}_V \circ \mathrm{Id}_V = \mathrm{Id}_V$ with $\mathcal{B}$ used at both ends. Hence $QP = I_n$. Reversing the order gives $PQ = I_n$. Thus $Q = P^{-1}$. □

Proposition 37 (Change-of-basis formula)**.** *Let $T \in \mathcal{L}(V, V)$, and let*

$$A = [T]^{\mathcal{B}}_{\mathcal{B}}, \qquad B = [T]^{\mathcal{C}}_{\mathcal{C}}.$$

With $P = [\mathrm{Id}_V]^{\mathcal{B}}_{\mathcal{C}}$, we have

$$B = PAP^{-1} = PAQ.$$

Proof. Applying T and then changing coordinates has the same effect as changing coordinates first and then applying T in the new basis. The following diagram expresses this equality:

$$\begin{array}{ccc} [v]_{\mathcal{B}} & \xrightarrow{A} & [T(v)]_{\mathcal{B}} \\ {\scriptstyle P}\downarrow & & \downarrow{\scriptstyle P} \\ [v]_{\mathcal{C}} & \xrightarrow{B} & [T(v)]_{\mathcal{C}}. \end{array}$$

Equivalently, the identity $T \circ \mathrm{Id}_V = \mathrm{Id}_V \circ T$ gives

$$BP = PA.$$

Multiplying on the right by $P^{-1} = Q$ yields $B = PAP^{-1} = PAQ$. □

To interpret the formula, start with a vector in $\mathcal{C}$-coordinates. The rightmost factor P^{-1} converts it to $\mathcal{B}$-coordinates, A computes the coordinates of its image under T, and P converts the result back to $\mathcal{C}$-coordinates.

16. September 30, 2026

16.1. **A change-of-basis example.** Recall Proposition 37 and its notation. If $\mathcal{B}$ and $\mathcal{C}$ are ordered bases of V, then

$$A = [T]_{\mathcal{B}}^{\mathcal{B}}, \qquad B = [T]_{\mathcal{C}}^{\mathcal{C}}, \qquad P = [\mathrm{Id}_V]_{\mathcal{C}}^{\mathcal{B}}$$

satisfy $B = PAP^{-1}$. Equivalently, if $Q = P^{-1} = [\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{C}}$, then

$$A = QBQ^{-1}, \qquad B = Q^{-1}AQ.$$

Thus the direction of the coordinate change determines which matrix appears on the left of A.

Example 34. *Let $V = \mathcal{P}_2(\mathbb{R})$, with ordered bases*

$$\mathcal{B} = (1, x, x^2), \qquad \mathcal{C} = (1, x+1, (x+1)^2),$$

and define $T\colon V \to V$ by $(Tp)(x) = xp'(x)$. Expanding the vectors of $\mathcal{C}$ in the basis $\mathcal{B}$ gives

$$Q = [\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{C}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \qquad P = Q^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since $T(1) = 0$, $T(x) = x$, and $T(x^2) = 2x^2$, we have

$$A = [T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Therefore

$$\begin{aligned} B = [T]_{\mathcal{C}}^{\mathcal{C}} &= Q^{-1}AQ \\ &= \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}.$$

We can also check the columns directly:

$$\begin{aligned} T(1) &= 0, \\ T(x+1) &= x = -1 + (x+1), \\ T((x+1)^2) &= 2x(x+1) = -2(x+1) + 2(x+1)^2. \end{aligned}$$

16.2. **Polynomials and their zeros.** We now turn to polynomials (Chapter 4 of Axler). Write $\mathcal{P}(\mathbb{F})$ for the space of polynomials over $\mathbb{F}$ and $\mathcal{P}_m(\mathbb{F})$ for those of degree at most m, together with the zero polynomial.

Definition 28 (Degree and zeros)**.** *A nonzero polynomial*

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m, \qquad a_0, \ldots, a_m \in \mathbb{F}, \quad a_m \neq 0,$$

has degree m*, denoted* $\deg p = m$*. We use the convention* $\deg 0 = -\infty$*. A scalar* $\lambda \in \mathbb{F}$ *is a* zero *or* root *of* p *if* $p(\lambda) = 0$*.*

Each polynomial defines a function $\mathbb{F} \to \mathbb{F}$ by evaluation. Over an infinite field, in particular $\mathbb{R}$ or $\mathbb{C}$, this function uniquely determines the coefficients, as we prove below. Over a finite field, we distinguish a polynomial from the function it defines.

Proposition 38 (Factor theorem)**.** *Let* $p \in \mathcal{P}(\mathbb{F})$ *have degree* $m \geq 1$*, and let* $\lambda \in \mathbb{F}$*. Then* $p(\lambda) = 0$ *if and only if there is a polynomial* q *of degree* $m - 1$ *such that*

$$p(z) = (z - \lambda) q(z).$$

Proof. If $p(z) = (z - \lambda)q(z)$, evaluation at $z = \lambda$ gives $p(\lambda) = 0$.

Conversely, suppose $p(\lambda) = 0$, and write $p(z) = a_0 + a_1 z + \cdots + a_m z^m$, where $a_m \neq 0$. Using

$$z^k - \lambda^k = (z - \lambda) \sum_{j=0}^{k-1} z^{k-1-j} \lambda^j \qquad (k \geq 1),$$

we obtain

$$\begin{aligned} p(z) &= p(z) - p(\lambda) \\ &= \sum_{k=1}^{m} a_k (z^k - \lambda^k) \\ &= (z-\lambda) \underbrace{\sum_{k=1}^{m} a_k \sum_{j=0}^{k-1} z^{k-1-j} \lambda^j}_{q(z)}. \end{aligned}$$

The leading term of q is $a_m z^{m-1}$, so $\deg q = m - 1$. □

Proposition 39. *A nonzero polynomial of degree m has at most m distinct roots in $\mathbb{F}$.*

Proof. We argue by induction on m. A nonzero constant has no roots. For $m = 1$, the polynomial $a_0 + a_1 z$, with $a_1 \neq 0$, has the unique root $-a_0/a_1$.

Suppose $m \geq 2$ and the result holds for degree $m - 1$. If p has no roots, there is nothing to prove. Otherwise, choose a root λ. By Proposition 38,

$$p(z) = (z - \lambda) q(z), \qquad \deg q = m - 1.$$

If $\mu \neq \lambda$ is another root of p, then $0 = (\mu - \lambda) q(\mu)$ implies $q(\mu) = 0$. By the induction hypothesis, q has at most $m - 1$ distinct roots. Including λ, the polynomial p therefore has at most m. □

Example 35. *The polynomial $p(z) = z^2 + 1$ has no roots in $\mathbb{R}$, but has exactly two roots in $\mathbb{C}$, namely i and $-i$.*

16.3. **Uniqueness of polynomial coefficients.**

Corollary 5. *Over an infinite field $\mathbb{F}$, a polynomial's values as a function $\mathbb{F} \to \mathbb{F}$ uniquely determine its coefficients.*

Proof. Suppose that, for every $z \in \mathbb{F}$,

$$a_0 + a_1 z + \cdots + a_m z^m = b_0 + b_1 z + \cdots + b_n z^n.$$

Without loss of generality, assume $m \geq n$, and put $b_j = 0$ for $n < j \leq m$. The polynomial

$$h(z) = \sum_{j=0}^{m} (a_j - b_j) z^j$$

vanishes at every element of $\mathbb{F}$, so it has infinitely many roots. Proposition 39 implies that h is the zero polynomial. Thus $a_j = b_j$ for every j. □

In particular, over an infinite field the degree of a nonzero polynomial function is well-defined.

Example 36 (Why the infinite-field hypothesis matters)**.** *Over $\mathbb{F}_2$, the nonzero polynomial $p(z) = z^2 + z$ defines the zero function: $p(0) = 0$ and $p(1) = 1 + 1 = 0$. Thus distinct polynomials can give the same function on a finite field. As a polynomial, $z^2 + z$ still has degree 2.*

16.4. **The division algorithm for polynomials.** For integers $a \geq 0$ and $b > 0$, the division algorithm gives unique integers $q, r \geq 0$ such that $a = qb + r$ and $0 \leq r < b$. For polynomials, degree takes the place of the size of an integer.

Theorem 3 (Polynomial division algorithm)**.** *Let $p, s \in \mathcal{P}(\mathbb{F})$, with $s \neq 0$. There exist unique polynomials $q, r \in \mathcal{P}(\mathbb{F})$ such that*

$$p = qs + r, \qquad \deg r < \deg s.$$

The polynomials q and r are the quotient *and* remainder*, respectively.*

Example 37. *If $p(z) = z^2 + 1$ and $s(z) = z - 2$, then*

$$z^2 + 1 = z(z-2) + 2(z-2) + 5 = (z+2)(z-2) + 5.$$

Hence $q(z) = z + 2$ and $r(z) = 5$.

Proof of Theorem 3. If $p = 0$, take $q = r = 0$. Otherwise, let $n = \deg p$ and $m = \deg s$. If $n < m$, take $q = 0$ and $r = p$.

Suppose $n \geq m$. Consider the following list in $\mathcal{P}_n(\mathbb{F})$:

$$1, z, \ldots, z^{m-1}, \; s, zs, \ldots, z^{n-m}s.$$

Its members have distinct degrees $0, 1, \dots, n$, with nonzero leading coefficients. They are linearly independent: in a nontrivial linear combination, the term of highest degree with a nonzero coefficient cannot be canceled by the others. There are $n+1 = \dim \mathcal{P}_n(\mathbb{F})$ members, so they form a basis. Therefore there are unique scalars such that

$$p(z) = \underbrace{r_0 + r_1 z + \cdots + r_{m-1} z^{m-1}}_{r(z)} + \underbrace{(q_0 + q_1 z + \cdots + q_{n-m} z^{n-m})}_{q(z)} s(z).$$

This gives $p = qs + r$ with $\deg r < m$. If $m = 0$, the initial list $1, z, \dots, z^{m-1}$ is empty, and $r = 0$.

To verify uniqueness in all cases, suppose $p = q_1 s + r_1 = q_2 s + r_2$, with $\deg r_1, \deg r_2 < m$. Then

$$(q_1 - q_2)s = r_2 - r_1.$$

If $q_1 - q_2 \neq 0$, the left side has degree $\deg(q_1 - q_2) + m \geq m$, whereas the right side has degree less than m, a contradiction. Hence $q_1 = q_2$ and $r_1 = r_2$. □

16.5. **Factorization over the complex and real numbers.**

Theorem 4 (Factorization over $\mathbb{C}$)**.** *Every nonconstant polynomial $p \in \mathcal{P}(\mathbb{C})$ of degree m has a factorization*

$$p(z) = c(z - \lambda_1) \cdots (z - \lambda_m),$$

where $c \in \mathbb{C} \setminus \{0\}$ and $\lambda_1, \dots, \lambda_m \in \mathbb{C}$. The scalar c is the leading coefficient of p, and the roots, listed with multiplicity, are unique up to permutation.

This is the factorization form of the fundamental theorem of algebra. Repeated roots are allowed; the root bound counts distinct roots.

Theorem 5 (Factorization over $\mathbb{R}$)**.** *Every nonconstant polynomial $p \in \mathcal{P}(\mathbb{R})$ has a factorization*

$$p(x) = c \prod_{j=1}^{k} (x - \lambda_j) \prod_{\ell=1}^{r} (x^2 + b_\ell x + c_\ell),$$

where $c \in \mathbb{R} \setminus \{0\}$, $\lambda_j, b_\ell, c_\ell \in \mathbb{R}$, *and*

$$b_\ell^2 < 4c_\ell \qquad (\ell = 1, \dots, r).$$

The factorization is unique up to reordering the factors, with multiplicities included, and $\deg p = k + 2r$. *Either product may be empty.*

The inequality $b_\ell^2 < 4c_\ell$ says that the quadratic factor has negative discriminant, so it has no real roots. Such factors arise by pairing nonreal conjugate roots: if $\lambda = a + bi$ with $b \neq 0$, then

$$(x - \lambda)(x - \overline{\lambda}) = x^2 - 2ax + (a^2 + b^2),$$

whose discriminant is $-4b^2 < 0$.

17. October 2, 2026

17.1. **Operators and invariant subspaces.**

Definition 29 (Linear operator)**.** *A linear map from a vector space* V *to itself is called a* linear operator *on* V. *We write* $\mathcal{L}(V) = \mathcal{L}(V, V)$ *for the space of all operators on* V.

One goal in studying an operator $T \in \mathcal{L}(V)$ is to find a direct sum decomposition

$$V = V_1 \oplus \cdots \oplus V_n$$

into subspaces that T preserves. We can then study T by studying its restrictions to these subspaces.

Definition 30 (Invariant subspace)**.** *Let* $T \in \mathcal{L}(V)$. *A subspace* $U \subseteq V$ *is* invariant under T *if* $T(u) \in U$ *for every* $u \in U$; *equivalently,* $T(U) \subseteq U$.

When U is invariant under T, the restriction $T|_U \colon U \to U$ is itself an operator on U.

Example 38. *Define* $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ *by* $D(p) = p''$. *The subspace* $\mathcal{P}_5(\mathbb{R})$ *is invariant under* D: *taking two derivatives of a polynomial of degree at most* 5 *produces a polynomial of degree at most* 3, *or the zero polynomial, and hence an element of* $\mathcal{P}_5(\mathbb{R})$.

Example 39. *For every $T \in \mathcal{L}(V)$, the subspaces*

$$\{0\}, \qquad V, \qquad \ker T, \qquad \operatorname{im} T$$

are invariant under T. The first two assertions follow from $T(0) = 0$ and the fact that T maps V to itself. If $u \in \ker T$, then $T(u) = 0 \in \ker T$. If $u \in \operatorname{im} T$, then $u = T(v)$ for some $v \in V$, and $T(u) = T(T(v)) \in \operatorname{im} T$.

17.2. **Eigenvalues and eigenvectors.** Consider a one-dimensional subspace of V. Let $v \in V \setminus \{0\}$ and $U = \operatorname{span}\{v\}$. If U is invariant under T, then $T(v) \in U$, so there is a scalar $\lambda \in \mathbb{F}$ such that

$$T(v) = \lambda v.$$

Conversely, if $T(v) = \lambda v$, then for every $a \in \mathbb{F}$,

$$T(av) = aT(v) = a\lambda v \in U.$$

Thus U is invariant under T precisely when $T(v)$ is a scalar multiple of v.

Definition 31 (Eigenvalue and eigenvector)**.** *Let $T \in \mathcal{L}(V)$. A scalar $\lambda \in \mathbb{F}$ is an* eigenvalue *of T if there exists a nonzero vector $v \in V$ such that $T(v) = \lambda v$. Such a vector v is called an* eigenvector *of T corresponding to λ.*

Consequently, V has a one-dimensional subspace invariant under T if and only if T has an eigenvalue.

Proposition 40. *Suppose that V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. The following are equivalent:*

(i) λ is an eigenvalue of T;
(ii) $T - \lambda \operatorname{Id}_V$ is not injective;
(iii) $T - \lambda \operatorname{Id}_V$ is not surjective;
(iv) $T - \lambda \operatorname{Id}_V$ is not invertible.

Proof. For $v \in V$,

$$T(v) = \lambda v \quad \iff \quad (T - \lambda \operatorname{Id}_V)(v) = 0.$$

Hence λ is an eigenvalue if and only if $\ker(T - \lambda \operatorname{Id}_V) \neq \{0\}$, which is equivalent to $T - \lambda \operatorname{Id}_V$ not being injective. This proves (i)$\Longleftrightarrow$(ii). Since $T - \lambda \operatorname{Id}_V$ is an operator on a finite-dimensional space, Proposition 32 gives the equivalence of (ii), (iii), and (iv). □

In particular, the eigenvectors corresponding to λ are exactly the nonzero vectors in $\ker(T - \lambda \operatorname{Id}_V)$. This kernel includes the zero vector, but the zero vector is never an eigenvector.

17.3. **A rotation over the real and complex numbers.**

Example 40. *Let $T \in \mathcal{L}(\mathbb{F}^2)$ be defined by*

$$T \begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} -z \\ w \end{bmatrix}.$$

Its matrix with respect to the standard basis is

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

If $\mathbb{F} = \mathbb{R}$, then T is counterclockwise rotation by $90°$. No nonzero real vector is taken to a scalar multiple of itself. Thus T has no real eigenvalues and no eigenvectors in $\mathbb{R}^2$.

To see algebraically how the field matters, the eigenvector equation is

$$\begin{bmatrix} -z \\ w \end{bmatrix} = \lambda \begin{bmatrix} w \\ z \end{bmatrix}, \qquad \text{so} \qquad \lambda w = -z, \quad \lambda z = w.$$

If $z = 0$, then $w = 0$, which is excluded for an eigenvector. Therefore $z \neq 0$, and substitution gives

$$\lambda^2 z = -z \quad \Longrightarrow \quad \lambda^2 = -1.$$

There is no solution in $\mathbb{R}$. Over $\mathbb{C}$, the only possibilities are $\lambda = i$ and $\lambda = -i$, and both occur:

$$T \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix} = i \begin{bmatrix} 1 \\ -i \end{bmatrix},$$

$$T \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i \\ 1 \end{bmatrix} = -i \begin{bmatrix} 1 \\ i \end{bmatrix}.$$

More precisely,

$$\ker(T - i\,\mathrm{Id}_{\mathbb{C}^2}) = \operatorname{span}\left\{\begin{bmatrix}1\\-i\end{bmatrix}\right\}, \qquad \ker(T + i\,\mathrm{Id}_{\mathbb{C}^2}) = \operatorname{span}\left\{\begin{bmatrix}1\\i\end{bmatrix}\right\}.$$

Thus the eigenvectors for i are $(w, -iw)$ with $w \neq 0$, and those for $-i$ are (w, iw) with $w \neq 0$.

17.4. **Eigenvectors corresponding to distinct eigenvalues.**

Proposition 41. *Let $T \in \mathcal{L}(V)$. If $v_1, \ldots, v_n$ are eigenvectors of T corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_n$, then $v_1, \ldots, v_n$ are linearly independent.*

Proof. Suppose, to the contrary, that the list is linearly dependent. Choose a dependent sublist of smallest length m and relabel it as $v_1, \ldots, v_m$. Denote their eigenvalues by $\lambda_1, \ldots, \lambda_m$. All eigenvectors are nonzero, so $m \geq 2$. There are scalars $a_1, \ldots, a_m$, not all zero, such that

$$a_1 v_1 + \cdots + a_m v_m = 0. \tag{2}$$

Minimality of m implies that every a_j is nonzero; otherwise, omitting a term would give a shorter dependent sublist. Applying T to (2) gives

$$a_1\lambda_1 v_1 + \cdots + a_m\lambda_m v_m = 0.$$

Subtracting λ_1 times (2), we obtain

$$a_2(\lambda_2 - \lambda_1)v_2 + \cdots + a_m(\lambda_m - \lambda_1)v_m = 0.$$

Each coefficient is nonzero because $a_j \neq 0$ and the eigenvalues are distinct. This is a nontrivial dependence relation among $m - 1$ of the eigenvectors, contradicting the minimality of m. □

Corollary 6. *Every operator on a finite-dimensional vector space V has at most $\dim V$ distinct eigenvalues.*

Proof. Choose one eigenvector for each distinct eigenvalue. By Proposition 41, any finite list of these vectors is linearly independent. Such a list has length at most $\dim V$, so there cannot be more than $\dim V$ distinct eigenvalues. □

18. October 5, 2026

Throughout, V is a vector space over $\mathbb{F}$, and $\mathcal{L}(V)$ denotes the space of linear operators on V. We use Id_V for the identity operator and $\mathcal{P}(\mathbb{F})$ for the space of polynomials over $\mathbb{F}$.

18.1. **The number of distinct eigenvalues.** Recall Corollary 6: an operator on a finite-dimensional vector space V has at most $\dim V$ distinct eigenvalues. Indeed, if $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of T and $v_1, \dots, v_m$ are corresponding eigenvectors, then Proposition 41 says that these vectors are linearly independent. Hence $m \le \dim V$.

18.2. **Powers of an operator.**

Definition 32 (Powers of an operator)**.** *Let $T \in \mathcal{L}(V)$. For a positive integer m, define*

$$T^m = \underbrace{T \circ \cdots \circ T}_{m \text{ times}}, \qquad T^0 = \mathrm{Id}_V .$$

If T is invertible, also define

$$T^{-m} = (T^{-1})^m \qquad (m \ge 1).$$

Composition of operators will often be written as multiplication: $ST = S \circ T$, so $(ST)(v) = S(T(v))$. Associativity of composition gives

$$T^m T^n = T^{m+n}, \qquad (T^m)^n = T^{mn}.$$

These identities hold for nonnegative integers m, n for every operator T, and for all integers m, n when T is invertible.

18.3. **Polynomials applied to operators.**

Definition 33 (Polynomial of an operator)**.** *If $T \in \mathcal{L}(V)$ and*

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m \in \mathcal{P}(\mathbb{F}),$$

define the operator

$$p(T) = a_0 \,\mathrm{Id}_V + a_1 T + \cdots + a_m T^m = \sum_{j=0}^{m} a_j T^j.$$

The constant term becomes a scalar multiple of the identity operator. Thus $p(T)$ is an operator on V, and $p(T)v$ is a vector in V.

Example 41 (A polynomial in the differentiation operator)**.** *Let $D \in \mathcal{L}(\mathcal{P}(\mathbb{F}))$ be the differentiation operator $Dq = q'$, and let $p(z) = 7 + z + z^2$. Then*

$$p(D) = 7\,\mathrm{Id}_{\mathcal{P}(\mathbb{F})} + D + D^2.$$

Consequently, for every $q \in \mathcal{P}(\mathbb{F})$,

$$p(D)q = 7q + Dq + D(Dq) = 7q + q' + q''.$$

For a fixed operator T, the map

$$\mathcal{P}(\mathbb{F}) \longrightarrow \mathcal{L}(V), \qquad p \longmapsto p(T),$$

is linear. Explicitly, for $a, b \in \mathbb{F}$ and polynomials p, q,

$$(ap + bq)(T) = ap(T) + bq(T).$$

This follows by adding coefficients of equal powers, inserting zero coefficients if the degrees differ.

The product of two polynomials is defined by $(pq)(z) = p(z)q(z)$. Polynomial evaluation at a fixed operator also preserves products.

Proposition 42. *Suppose $p, q \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$. Then*

(i) $(pq)(T) = p(T)q(T)$*;*
(ii) $p(T)q(T) = q(T)p(T)$*.*

Proof (left as an exercise in lecture). Write

$$p(z) = \sum_{j=0}^{m} a_j z^j, \qquad q(z) = \sum_{k=0}^{n} b_k z^k.$$

Distributivity and the power law give

$$p(T)q(T) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k T^{j+k} = (pq)(T).$$

Since $pq = qp$, the first identity also yields

$$p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T). \qquad \square$$

Remark 1. Arbitrary operators need not commute. The proposition says that polynomials in the same operator commute. In particular, taking $q(z) = z$ shows that $p(T)T = Tp(T)$.

18.4. **Invariant subspaces from polynomials.**

Proposition 43. *If $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbb{F})$, then $\ker p(T)$ and $\operatorname{im} p(T)$ are invariant under T.*

Proof. First, let $u \in \ker p(T)$, so $p(T)u = 0$. Because T commutes with $p(T)$,

$$p(T)(Tu) = T(p(T)u) = T(0) = 0.$$

Thus $Tu \in \ker p(T)$, proving invariance of the kernel.

Next, let $u \in \operatorname{im} p(T)$. There is a vector $v \in V$ such that $u = p(T)v$. Again using commutativity,

$$Tu = T(p(T)v) = p(T)(Tv) \in \operatorname{im} p(T).$$

Thus the image is invariant as well. □

18.5. **Existence of eigenvalues over the complex numbers.**

Theorem 6. *Every operator on a nonzero finite-dimensional complex vector space has an eigenvalue.*

Proof. Let $T \in \mathcal{L}(V)$, where V is a complex vector space of dimension $n > 0$. Choose $v \in V \setminus \{0\}$. The list

$$v, Tv, T^2v, \ldots, T^nv$$

contains $n + 1$ vectors in an n-dimensional space, so it is linearly dependent. Thus there are scalars $a_0, \ldots, a_n$, not all zero, such that

$$a_0v + a_1Tv + \cdots + a_nT^nv = 0.$$

Equivalently, some nonzero polynomial p satisfies $p(T)v = 0$. Choose such a polynomial of smallest degree. Its degree is at least one, because a nonzero constant polynomial gives $p(T)v = a_0v \neq 0$.

By the fundamental theorem of algebra (Theorem 4), p has a root $\lambda \in \mathbb{C}$. The factor theorem gives

$$p(z) = (z - \lambda)q(z), \qquad \deg q = \deg p - 1.$$

By Proposition 42,

$$p(T) = (T - \lambda \operatorname{Id}_V)q(T).$$

Minimality of the degree of p implies $q(T)v \neq 0$, since q is a nonzero polynomial of smaller degree. Setting $w = q(T)v$, we therefore have

$$w \neq 0, \qquad (T - \lambda \operatorname{Id}_V)w = p(T)v = 0.$$

Hence $Tw = \lambda w$, so λ is an eigenvalue of T with eigenvector w. □

Remark 2 (The minimal-degree polynomial used in the proof). The polynomial above is chosen to satisfy $p(T)v = 0$ for one fixed nonzero vector v. This does not assert $p(T) = 0$ as an operator on all of V. The minimal polynomial of an operator instead concerns an identity on the whole space. The eigenvalue argument here needs only the relation for v.

18.6. **Why the hypotheses matter.**

Example 42 (A real operator without eigenvalues)**.** *Recall the rotation $T(x, y) = (-y, x)$ on $\mathbb{R}^2$ from Example 40 has no real eigenvalues. Indeed, $T^2 = -\operatorname{Id}_{\mathbb{R}^2}$, so $Tv = \lambda v$ for a nonzero real vector would imply*

$$-v = T^2 v = \lambda^2 v, \qquad \lambda^2 = -1,$$

which is impossible for $\lambda \in \mathbb{R}$.

Example 43 (An operator on an infinite-dimensional space)**.** *Consider the complex vector space $\mathcal{P}(\mathbb{C})$, which is infinite-dimensional. Define the multiplication operator*

$$T \in \mathcal{L}(\mathcal{P}(\mathbb{C})), \qquad (Tp)(z) = zp(z).$$

If $p \neq 0$, then $\deg(Tp) = \deg p + 1$. For $\lambda \neq 0$, the polynomial λp has degree $\deg p$, so Tp cannot equal λp. For $\lambda = 0$, the equation is also impossible because $Tp \neq 0$. Thus T has no eigenvalues.

19. October 7, 2026

Throughout, V is a finite-dimensional vector space over $\mathbb{F}$, and $T \in \mathcal{L}(V)$. We continue to use $\mathcal{P}(\mathbb{F})$ for the space of polynomials over $\mathbb{F}$.

19.1. **Monic polynomials and annihilating polynomials.**

Definition 34 (Monic polynomial)**.** *A nonzero polynomial is* monic *if its leading coefficient is* 1*. Thus a monic polynomial of degree m has the form*

$$p(z) = c_0 + c_1 z + \cdots + c_{m-1} z^{m-1} + z^m.$$

In particular, the constant polynomial 1 *is monic of degree* 0*.*

Recall that $p(T) = 0$ means that $p(T)$ is the zero operator: $p(T)v = 0$ for every $v \in V$. A polynomial satisfying this identity is said to *annihilate* T.

Proposition 44. *There is a unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$. Furthermore,*

$$\deg p \leq \dim V.$$

Proof. We first prove by induction on $\dim V$ that there is a monic polynomial of degree at most $\dim V$ that annihilates T. If $\dim V = 0$, then Id_V is the zero operator, so the constant polynomial $p = 1$ works.

Suppose now that $n = \dim V > 0$, and assume the existence statement holds for operators on spaces of dimension less than n. Choose $u \in V \setminus \{0\}$. The list

$$u, Tu, T^2u, \ldots, T^n u$$

is linearly dependent because it contains $n + 1$ vectors. Let m be the smallest positive integer such that $T^m u$ is a linear combination of $u, Tu, \ldots, T^{m-1}u$. Then $1 \leq m \leq n$, the list $u, Tu, \ldots, T^{m-1}u$ is linearly independent, and there are scalars $c_0, \ldots, c_{m-1}$ such that

$$c_0 u + c_1 Tu + \cdots + c_{m-1} T^{m-1} u + T^m u = 0.$$

Define the monic polynomial

$$q(z) = c_0 + c_1 z + \cdots + c_{m-1} z^{m-1} + z^m.$$

Thus $q(T)u = 0$. By Proposition 42, $q(T)$ commutes with every power of T. Hence, for every $k \geq 0$,

$$q(T)(T^k u) = T^k(q(T)u) = 0.$$

In particular, $\ker q(T)$ contains the m linearly independent vectors $u, Tu, \ldots, T^{m-1}u$. Therefore

$$\dim \ker q(T) \geq m.$$

The fundamental theorem of linear maps (Theorem 2) now gives

$$\dim \operatorname{im} q(T) = n - \dim \ker q(T) \leq n - m < n.$$

Set $U = \operatorname{im} q(T)$. By Proposition 43, U is invariant under T, so $T|_U \in \mathcal{L}(U)$. We may apply the induction hypothesis to obtain a monic polynomial s with

$$\deg s \leq \dim U \leq n - m, \qquad s(T|_U) = 0.$$

Since U is invariant under T, polynomial evaluation commutes with restriction: $s(T)|_U = s(T|_U)$. For every $v \in V$, the vector $q(T)v$ belongs to U, and thus

$$(sq)(T)v = s(T)q(T)v = s(T|_U)(q(T)v) = 0.$$

Consequently, sq annihilates T. It is monic, and

$$\deg(sq) = \deg s + \deg q \leq (n - m) + m = n.$$

This completes the induction.

Among all monic polynomials annihilating T, choose one, p, of smallest degree. The existence argument implies $\deg p \leq \dim V$. To prove uniqueness, suppose r is another monic polynomial of the same smallest degree with $r(T) = 0$. Then

$$(p - r)(T) = 0, \qquad \deg(p - r) < \deg p,$$

because the leading terms cancel. If $p - r \neq 0$, dividing $p - r$ by its leading coefficient produces a monic polynomial of smaller degree that still annihilates T, a contradiction. Therefore $p = r$. □

19.2. **The minimal polynomial.**

Definition 35 (Minimal polynomial)**.** *The* minimal polynomial *of T is the unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$.*

If $V = \{0\}$, the minimal polynomial is 1. If $V \neq \{0\}$, then no nonzero constant polynomial annihilates T, because $c \operatorname{Id}_V \neq 0$ for $c \neq 0$. Thus in this case

$$1 \leq \deg p \leq \dim V.$$

19.3. **Computing the minimal polynomial using matrices.** Suppose $n = \dim V > 0$. We seek the smallest positive integer m for which there are scalars $c_0, \ldots, c_{m-1}$ satisfying

$$c_0 \operatorname{Id}_V + c_1 T + \cdots + c_{m-1} T^{m-1} = -T^m.$$

The corresponding minimal polynomial is then

$$p(z) = c_0 + c_1 z + \cdots + c_{m-1} z^{m-1} + z^m.$$

Choose an ordered basis $\mathcal{B}$ of V and put $A = [T]_{\mathcal{B}}^{\mathcal{B}}$. Since the matrix representation preserves sums, scalar multiplication, and composition, the operator equation is equivalent to

$$c_0 I_n + c_1 A + \cdots + c_{m-1} A^{m-1} = -A^m.$$

Equating corresponding matrix entries gives n^2 linear equations in the m unknowns $c_0, \ldots, c_{m-1}$. Try $m = 1, 2, \ldots, n$ and stop at the first value for which the system has a solution. Proposition 44 guarantees that such a value occurs by $m = n$.

19.4. **A shortcut using a single vector.** A faster method sometimes works by applying the operator equation to just one vector. Choose $v \in V \setminus \{0\}$ and consider

$$c_0 v + c_1 T v + \cdots + c_{n-1} T^{n-1} v = -T^n v.$$

This gives only n scalar equations. The key condition is that

$$v, Tv, \ldots, T^{n-1} v$$

be a basis of V. Such a vector is called a *cyclic vector* for T.

If this condition holds, the equation determines unique coefficients $c_0, \dots, c_{n-1}$. Set

$$p(z) = c_0 + c_1 z + \cdots + c_{n-1} z^{n-1} + z^n.$$

Then $p(T)v = 0$, and commutativity gives

$$p(T)(T^k v) = T^k(p(T)v) = 0 \qquad (0 \le k \le n-1).$$

Thus $p(T)$ vanishes on a basis, so $p(T) = 0$. Moreover, no nonzero polynomial of degree less than n can even annihilate v: such a relation would contradict the linear independence of $v, Tv, \dots, T^{n-1}v$. Hence p is the minimal polynomial of T.

Remark 3. For an arbitrary nonzero vector v, the equation $p(T)v = 0$ need not imply $p(T) = 0$. The basis condition above is what makes this shortcut valid. If the list fails to be a basis, a polynomial found using v must still be checked on all of V.

Example 44. *Let $T \in \mathcal{L}(\mathbb{F}^5)$ have the following matrix with respect to the standard basis $\mathcal{E} = (e_1, \dots, e_5)$:*

$$A = [T]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} 0 & 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & 6 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Take $v = e_1$. Reading the columns of A, we obtain

$$\begin{aligned} v &= e_1, & Tv &= e_2, & T^2 v &= e_3, \\ T^3 v &= e_4, & T^4 v &= e_5, & T^5 v &= -3e_1 + 6e_2. \end{aligned}$$

The first five vectors are the standard basis, so v is cyclic. The equation

$$c_0 v + c_1 Tv + c_2 T^2 v + c_3 T^3 v + c_4 T^4 v = -T^5 v$$

becomes

$$c_0 e_1 + c_1 e_2 + c_2 e_3 + c_3 e_4 + c_4 e_5 = 3e_1 - 6e_2.$$

Hence

$$c_0 = 3, \qquad c_1 = -6, \qquad c_2 = c_3 = c_4 = 0.$$

Therefore the minimal polynomial of T *is*

$$\boxed{p(z) = z^5 - 6z + 3}.$$

In particular, $T^5 - 6T + 3\,\mathrm{Id}_{\mathbb{F}^5} = 0$*, and no nonzero polynomial of smaller degree annihilates* T*.*